

COLOURSPACE CMS

FULL PROFILING AND 3D LUT GENERATION GUIDE FOR LG OLED



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Introduction

With the addition of internal LUT capabilities on LG OLED consumer TV's back in 2018, it is extremely simple to achieve reference level picture quality in SDR, provided you follow some fundamental basic steps correctly.

This guide will attempt to outline those steps in an easy-to-follow manner, allowing even the most inexperienced user of ColourSpace CMS to achieve outstanding results in the shortest time possible.

ColourSpace CMS is the latest software from [Light Illusion](#), re-written from the ground up to replace LightSpace CMS to allow for future upgrade possibilities. The Light Illusion website has far more comprehensive guides than this one will be, and contains substantial amounts of information on all things calibration related. This guide can be seen as a simplified version as it will relate to best practices for LG OLED only using the displays internal LUT capabilities.

ColourSpace CMS is an extremely advanced calibration software with unique features not currently found in any other commercially available calibration software solution. This guide will not touch on many of these features. For those that are interested in seeing more in-depth explanations of some these, there are several videos and highly detailed guides on the Light Illusion website. This guide will contain links to web pages that may contain any relevant information.

The methods described here-after were born out of countless days, weeks and months of intensive testing performed by a very dedicated group of enthusiasts in varying environmental conditions with all levels of equipment, both professional reference level and consumer grade measurement probes and patch generators. The methods have also been verified to be consistently repeatable and reliable across model years of the LG OLED, from the initial 2018 models that were released with internal LUT upload capability, to the latest 2021 models. Therefore, the steps outlined can be trusted to produce excellent results each time.

The procedure outlined in this guide is continuously tested, checked and verified on a regular basis to ensure that it remains the best possible solution for LG OLED calibration when using the displays internal LUT capabilities. This means that the guide can be updated as and when necessary, in addition to verifying that when new models of LG OLED are released it will apply to them also.

Equipment used and referenced in the guide will be an X-Rite i1 Display Pro OEM (Rev.B) which is profiled to an X-Rite i1 Pro2 (Rev.E), along with the [Raspberry Pi based PGenerator](#). Most home enthusiast calibrators will be using this equipment, so all provided screenshots will be easier to understand if the same equipment is referenced in the guide.

Note: Since 2019, LG have provided an internal test pattern generator for calibration purposes. We do not recommend using this feature for display profiling and LUT generation due to stability and accuracy issues. For SDR display profiling, an external test pattern generator is recommended. No support will be given to users of the "iTPG" when calibrating for SDR as there are too many variables to account for. For home enthusiasts, the Raspberry Pi based PGenerator is the best accurate external TPG solution available.

Note: The i1 Display Pro Retail/Pro Plus Retail meters are not officially supported in ColourSpace, but can be "unlocked" to work with the software, as can any other "branded" i1 Display Pro based meter (e.g. SpectraCal C6 Probes). Please Contact Ted Aspiotis via his contact form at his website for more information.

https://displaycalibrations.com/contact_us.html

The target of the calibration will be to a reference Rec709 colour space, with a Power Law Gamma of 2.4 and a peak luminance of 100 nits (@100% White). Alternate white points, luminance targets and/or alternate gamma values are of course possible, but for the sake of simplicity the target for the guide will be to the reference standard.

Throughout this guide you will see references to 109% white. The reason for this is because the internal LUT container of LG OLED is designed to work in Video Extended range with TV's Contrast set to its default value of 85 to preserve super-white (109%). We will be profiling the display in 8-Bit resolution, Video Extended range (16-255). With 100% white measuring 100 nits, 109% white will be approximately 124 nits.

Pre-Calibration

Pre-Calibration Steps:

Pre-Calibration set-up is probably the single most important stage of the calibration process. This cannot be emphasised enough. If you get the initial set-up right, the rest of the process should go smoothly without issues.

Ted Aspiotis (ConnectEDDD @ AVSForum) has posted tremendous amounts of detail on much of the most important things. You can see this at the links provided here:

- [LG OLED Preparation before the profiling using ColourSpace](#)
- [LG OLED Troubleshooting](#)
- [Calibration notes for LG OLED 2020 TVs](#)
- [LG OLED Templates \(for LightSpace/ColourSpace users\)](#)

It is highly advised to spend some time reading through as much of the information provided within those posts as possible (and the additional links Ted has provided) before attempting to calibrate any LG OLED with a 3D LUT. It will help provide some understanding as to why some of the steps and settings in this guide are used, and will give the best chance of spectacular results each and every time.

Warm up your equipment:

No, don't wrap it all up snug in a blanket. Just power everything on. The TV should be powered on and playing standard content of some type for a minimum of 45 minutes (preferably an hour) to allow the displays electronic components to warm up and settle. The probe/probes should be powered on by connecting them to the USB port of your calibration computer for a minimum of 20-30 minutes too. Just powering them via USB is enough to warm them, there is no need to place them in their measurement position at this point.

Disable all processing:

If you have read through the links provided above then you will know which settings to disable, but I will list them here.

First, select the picture mode to be calibrated, such as ISF Expert (Dark Room). Once selected, check the following settings within the TV's main "All Settings"

Global Picture settings:

- | | | |
|-------------------------|------|--|
| • Aspect Ratio | 16:9 | <i>Set Just Scan – On</i> |
| • Energy Saving | Off | |
| • Eye Comfort Mode | Off | <i>Under Additional Settings</i> |
| • HDMI Deep Colour | On | <i>Under Additional Settings</i> |
| • Instant Game Response | Off | <i>Under Additional Settings</i> |
| • Screen Shift | Off | <i>Under Additional Settings -> OLED Panel Settings</i> |
| • Logo Luminance | Off | <i>Under Additional Settings -> OLED Panel Settings</i> |

Picture Mode Settings:

- | | | |
|------------------------|-----|--|
| • Sharpness | 0 | <i>Default value of 10 actually adds some sharpness processing</i> |
| • Dynamic Contrast | Off | <i>Under Expert Controls</i> |
| • Super Resolution | Off | <i>Under Expert Controls</i> |
| • Peak Brightness | Off | <i>Under Expert Controls</i> |
| • Noise Reduction | Off | <i>Under Picture Options</i> |
| • MPEG Noise Reduction | Off | <i>Under Picture Options</i> |
| • Smooth Gradation | Off | <i>Under Picture Options</i> |
| • Black Level | Low | <i>Under Picture Options</i> |
| • Motion Eye Care | Off | <i>Under Picture Options</i> |
| • TruMotion | Off | <i>Under Picture Options</i> |

General Settings:

- | | | |
|-------------------------|-----|---|
| • Eco Mode | Off | <i>2018 and 2019 LG OLED's only</i> |
| • Auto Power Off | Off | <i>Under General/Additional Settings/Eco Mode – 2020 LG OLED's Only</i> |
| • Sleep Timer | Off | <i>Under General/Time – 2020 LG OLED's Only</i> |
| • Power On/Off Timer | Off | <i>Under General/Time – 2020 LG OLED's Only</i> |
| • AI Picture Pro | Off | <i>AI Service – 2020 LG OLEDs only</i> |
| • AI Brightness Control | Off | <i>AI Service – 2020 LG OLEDs only</i> |
| • Auto Genre Selection | Off | <i>AI Service – 2020 LG OLEDs only</i> |

These options can all be disabled during the TV warm up period to save some time, but remember that each picture mode (ISF Dark, ISF Bright etc...) and each input (HDMI 1, HDMI 2 etc....) will need to have all these settings disabled independently. What this means is that you may have to momentarily switch to the HDMI input that your PGenerator (or other external TPG) is connected to, select the picture mode to be calibrated and disable all of the processing options listed above, then switch back to the source playing your content to continue display warm up.

Disable all processing (2021 LG OLEDs):

With the release of 2021 LG OLED's there has been a complete redesign of WebOS and its menu structure. The naming convention for several of its settings within those menus have also changed, and/or moved to other sub menus. This requires extra care to be taken to disable the settings listed above, as they may be named differently or located in other sub menus.

First, select the picture mode to be calibrated, such as ISF Expert (Dark Space, night). Once selected, check the following settings within the TV's main "All Settings"

Picture Settings:

- Aspect Ratio 16:9 *Set Just Scan – On*

Additional Settings:

- Reduce Blue Light Off

Advanced Settings:

- Auto Dynamic Contrast Off *Under Brightness*
- Peak Brightness Off *Under Brightness*
- Black Level Limited *Under Brightness*
- Motion Eye Care Off *Under Brightness*
- Colour Upgrade Off *Under Colour -> Fine Tune*
- Sharpness 0 *Under Clarity*
- Super Resolution Off *Under Clarity*
- Noise Reduction Off *Under Clarity*
- MPEG Noise Reduction Off *Under Clarity*
- Smooth Gradation Off *Under Clarity*
- TruMotion Off *Under Clarity*

General:

- AI Picture Pro Disabled *Under AI Service*
- AI Brightness Settings Disabled *Under AI Service*
- HDMI Deep Colour 4K *Under Devices -> HDMI Settings*

Support:

- Screen Move Off *Under Support -> OLED Screen Saver*
- Adjust Logo Brightness Off *Under Support -> OLED Screen Saver*
- Energy Saving Step Off *Under Support -> Energy Saving*
- Auto Power Off Disabled *Under Support -> Energy Saving*

It should be noted by 2021 LG OLED owners, that the procedure to calibrate from this point onwards is identical to that from all previous years. However, several other options and settings within the menu structure of the 2021 LG OLED's have been renamed also. These should all be self-explanatory, such as OLED Pixel Brightness (previously OLED Light) and Screen Brightness (previously Brightness).

Please take this into account when following the procedure laid out below.

Pre-Calibration Adjustments - Checks:

Reset the Picture Mode with a Unity LUT:

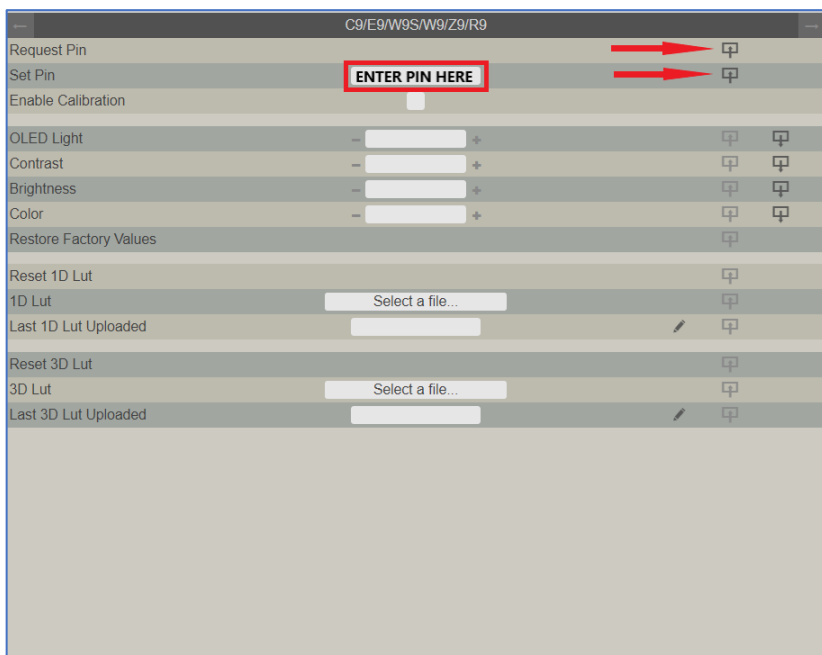
LG OLED's have issues when sending a Unity 1D LUT or Custom 1D LUT. The issues vary dependant on picture mode (SDR/HDR/Dolby Vision). For this reason, it is advised that you do not ever Reset the 1D LUT in the TV and calibrate with a 3D LUT only.

To Reset the LUT Container of the target picture mode, Device Control Template will be used. Launch Device Control, open your web-browser and navigate to <http://127.0.0.1:81/>

Select the correct Template for your specific TV model/year and connect to the Template.

Once connected to the Template, you will be required to send specific commands to the TV at various stages. Each command will need to be sent individually by clicking the "Send" button to right of each value or entry field. 

With the correct Template connected, send the command to "Request Pin"

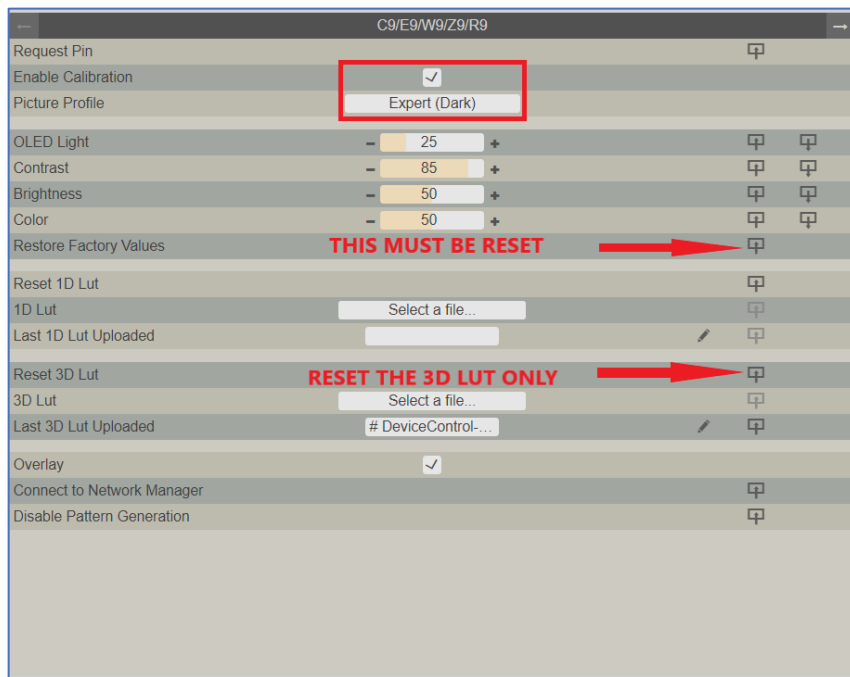


C9/E9/W9S/W9/Z9/R9	
Request Pin	Send
Set Pin	Send
Enable Calibration	☐
OLED Light	Send
Contrast	Send
Brightness	Send
Color	Send
Restore Factory Values	Send
Reset 1D Lut	Send
1D Lut	Send
Last 1D Lut Uploaded	Send
Reset 3D Lut	Send
3D Lut	Send
Last 3D Lut Uploaded	Send

Enter the PIN displayed on the LG OLED display into the "Set Pin" entry field and click to send. You will notice some pop-up messages on the screen of the LG confirming that Device Control is connecting to the TV.

Enable Calibration by placing a checkmark into the entry field. By default, the LG will *always* switch your TV into "Cinema" mode so click the Picture Profile entry field and scroll to Select "Expert (Dark)" to automatically switch the TV to the correct target picture mode to be calibrated.

Send the commands to “Restore Factory Values”, followed by “Reset 3D LUT”. Both of these are essential.



Each time you “Enable Calibration”, the TV will automatically switch to the Cinema picture mode. This is the way the TV works and is expected. Please ensure to select the desired target picture mode any time Calibration is enabled from within Device Control.

Note. It is not critical to do this here as it can be done in the User Menu controls, but you can set “OLED Light” to a value of approximately 26 for 100 nits peak white (this value can vary due to “Panel Variance”) by using the -/+ buttons and sending it to the TV.

2018 and 2020 LG OLED owners can Disable Calibration at this point, however 2019 LG OLED owners may find it beneficial to keep Calibration Enabled temporarily if they plan to Pre-Calibrate White in the Service Menu.

Check User Menu Settings:

Most of the settings may already be correct disabled if you did this at an earlier stage ([see above](#)), but it is worth checking that they have remained disabled as there are two more settings (and one optional) that need to be checked after the loading of the Unity 3D LUT.

- | | | |
|------------------------------|--|---|
| • Colour Gamut | Locked to Wide | <i>Under Picture Mode Settings -> Expert Controls</i> |
| • Colour Gamut (2021) | Locked to Native | <i>Under Picture -> Advanced Settings -> Colour</i> |
| • Gamma | 2.2 | <i>Under Picture Mode Settings -> Expert Controls</i> |
| • Gamma (2021) | 2.2 | <i>Under Picture -> Advanced Settings -> Brightness</i> |
| • White Balance | <i>Optional change explained in the Pre-Calibrating White guides (see below)</i> | |

Colour Gamut must be set to Wide/Native depending on model year (it should also be greyed out/locked) as you will be profiling the panel in its native gamut.

Gamma should be set at 2.2 (but it shouldn’t be locked). This is the panels processed native gamma. Altering this to another value will introduce further processing which is not desired or necessary when performing a 3D LUT calibration.

If Gamma is greyed out and locked at 2.2 along with White Balance Controls, then the 1D LUT has been previously reset. Only a “Restore to Initial Settings” can resolve this

Pre-Calibrate White:

If you are not confident with entering the Service Menu, or do not have access to a remote that will gain you entry, then you can skip this step.

For best results, a pre-calibrated peak white is desired. It is not essential, but by pre-calibrating the white point (in this case, to D65) at its peak value, it will help prevent luminance loss during the LUT generation. To do this, you will need to access the Service Menu of the LG.

Once again, Ted Aspiotis has posted detailed information on the process many times both at AVSForum and his own website.

- [Peak White Pre-Calibration Tip - Before Starting the 3D LUT profiling](#)
- [Calibration Notes](#) (approximately half way down the page)

Please take the time to fully read and understand the procedure before attempting to access the Service Menu. Any incorrect change of a critical setting in the Service Menu has the potential to “Brick” your TV.

There are several ways to enter the Service Menu:

- LG Service Remote *A web search for LG Service Remote MKJ39170828 will provide many options*
- Harmony Remote *Most models have the option provided after adding your LG as a device*
- [Sure Universal Remote](#) *Android app (iOS version has no SM access command at this time)*
- [LG Service Remote App](#) *Android app (Requires IR Transmitter on your Android device)*

There may be other smartphone apps available, but over the years some of these have either been completely removed from app stores, or had functionality severely reduced. The LG Service Remote and Harmony Remote methods are the most reliable way to access the Service Menu.

To enter the Service Menu to adjust White Balance, look for the EZ-Adjust button. On some Harmony Remotes it is actually named “Service Menu”, but it is more commonly called EZ-Adjust.

Never press the IN-STOP button. It will wipe your TV to factory settings and clear the usage counter which is critical for the “Pixel Refresher” to function correctly.

Users with a Spectroradiometer/photometer of some kind should use this probe to Pre-Calibrate White for ultimate accuracy.

Users without a Spectroradiometer/photometer should use the correct meter pre-set mode for their probe. For i1 Display Pro users, you should select WRGB OLED.

Special notice for 2019 LG OLED users only:

Adjustment of the Service Menu OLED Light control (*named Backlight in the Service Menu*) does not affect externally generated test patterns when setting the Service Menu “Test Pattern” to OFF. This unfortunately requires a slight workaround to the method described in Ted’s instructions.

- Leave “*Enable Calibration*” checked in Device Control Template so that you can use the controls there to make adjustments to raise/lower OLED Light to your target peak value as required.
- Enter the Service Menu and select the Colour Temperature to be adjusted
- Set Test Patterns to 100 IRE
- Set Backlight to the same value you have made in the User Menu or Device Control (e.g. OLED Light 26)
- Make adjustments targeting D65 at 100 Nits
- Press “Back” on your LG Magic Remote to display the externally generated patch
- Check measurements are now D65 and approximately 123 Nits, as the external patch is now 109%
- Raise or lower the OLED Light value with Device Control if required to reach approximately 123 Nits
- Make final adjustments to D65 if required

It may be necessary to repeat going in and out of the White Balance menu a few times to achieve this, but it should only take a few minutes to complete accurate Pre-Calibration of the White.

While in the Service Menu White Balance adjustment menu, DO NOT PRESS RESET

Once complete, please remember to “Disable Calibration” in Device Control Template

ASBL – TPC:

With access to the Service Menus, you also have the option to the displays Anti-Static Brightness Limiter (ASBL). This is a protection feature that notices when a static image is displayed for prolonged periods and gradually dims the panels light output to help prevent Burn-In. Unfortunately, the algorithm isn’t fool proof and often takes an already dim static image (such as low light patches) and dims them further. This can be seen with real content at times too, so many owners wish to disable the function.

For calibration profiling, it is preferable to have it disabled, but it can be enabled again after calibration is complete if you so desire.

To disable ASBL, enter the IN-START Menu and scroll to the option “**OLED**” and select it. Look to the right side of the screen to see **TPC Enable – On**. Toggle this setting to “**Off**” to disable ASBL.

GSR – Global Dimming

LG added the ability to fully disable all protective dimming features on consumer OLED TV’s. The feature is named GSR within the IN-START Menu. By default, it will read **GSR: Enable – On**. This function can now be fully disabled if so desired. It can be found at the bottom of the same menu as TPC. Toggle this setting to “**Off**” to disable GSR.

Both ASBL (TPC) and GSR are Burn-In protection features. Switching either or both of these off are done at your own risk!!!

Do not make any further adjustments to any other setting in either the EZ-Adjust, or IN-START Service Menus. Doing so could potentially brick your TV and/or void your warranty!!!!

Brightness and Contrast:

Using verified accurate patterns, Brightness and Contrast should be checked and adjusted if required.

As a rule, Brightness and Contrast settings on 50 and 85 (factory default values) *should* be correct on LG OLED's. But that is not guaranteed and can vary by panel. In addition to panel variation, viewing environment can have an effect also.

Do not be surprised if you have to raise Brightness to 55 or even higher. It has been known on some panels to require a Brightness setting over 60. This is panel variation at play

Ted's LightSpace CMS Calibration Disk is favoured by many to make these adjustments. It also has some exceptionally useful Colour Reproduction patterns for Post-Calibration verification.

See the [Ted's LightSpace CMS Calibration Disk](#) thread on AVSForum for detailed information and support using the disk. Also Available in digital format (MVK and MP4 versions).

It is now also available for use directly from the Raspberry Pi PGenerator. See [this post in the Dedicated Raspberry Pi PGenerator Thread](#) on AVSForum

Equipment setup in ColourSpace:

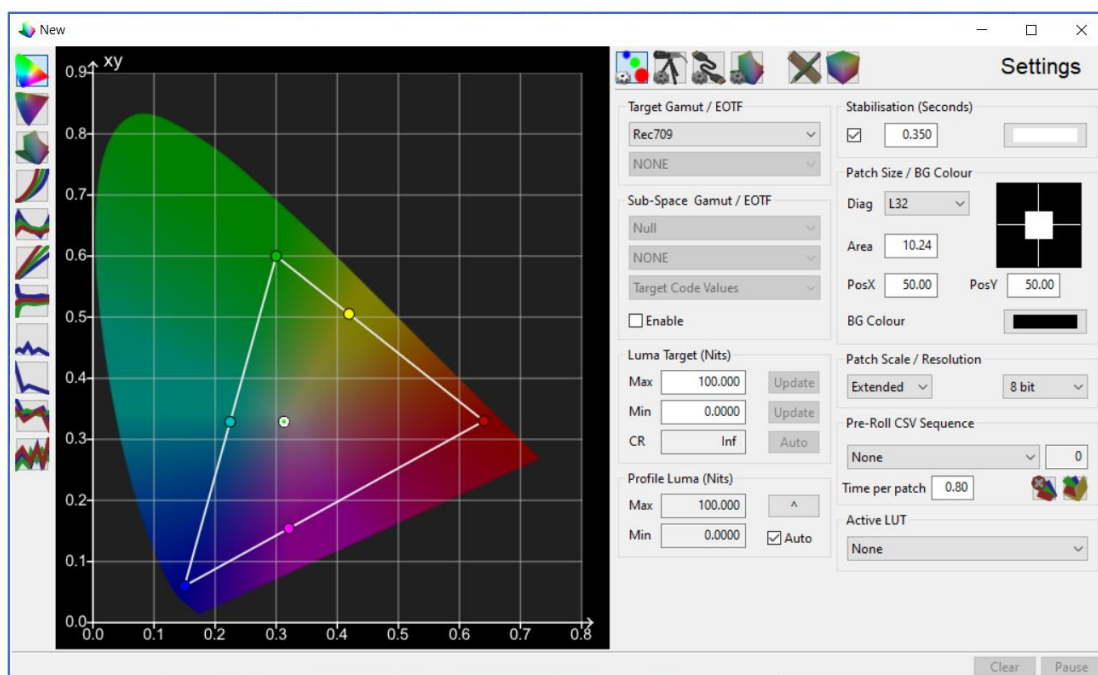
This may already have been done, especially if you have Pre-Calibrated White in the Service Menu. But let's run through it anyway.

Before commencing any further, it is advised to power off any external devices that are connected via HDMI and place the remotes for them out of reach. Accidental activation of an external device will cause a pop-up notification that it is available, or worse, automatically switch to that input. This will invalidate the profile.

Launch ColourSpace CMS and open a Profiling Window



When the Window first opens you will be presented with the main Settings tab



As we intend to profile up to 109%, we must set Patch Scale to Extended, which represents 16-255. Please ensure that Resolution is set to 8 bit, as the custom patch sequences are created with 8 bit RGB Triplets.

For this calibration we will be targeting a reference Rec709 target gamut which has a default power law gamma of 2.4. This is set in the top left of the settings.

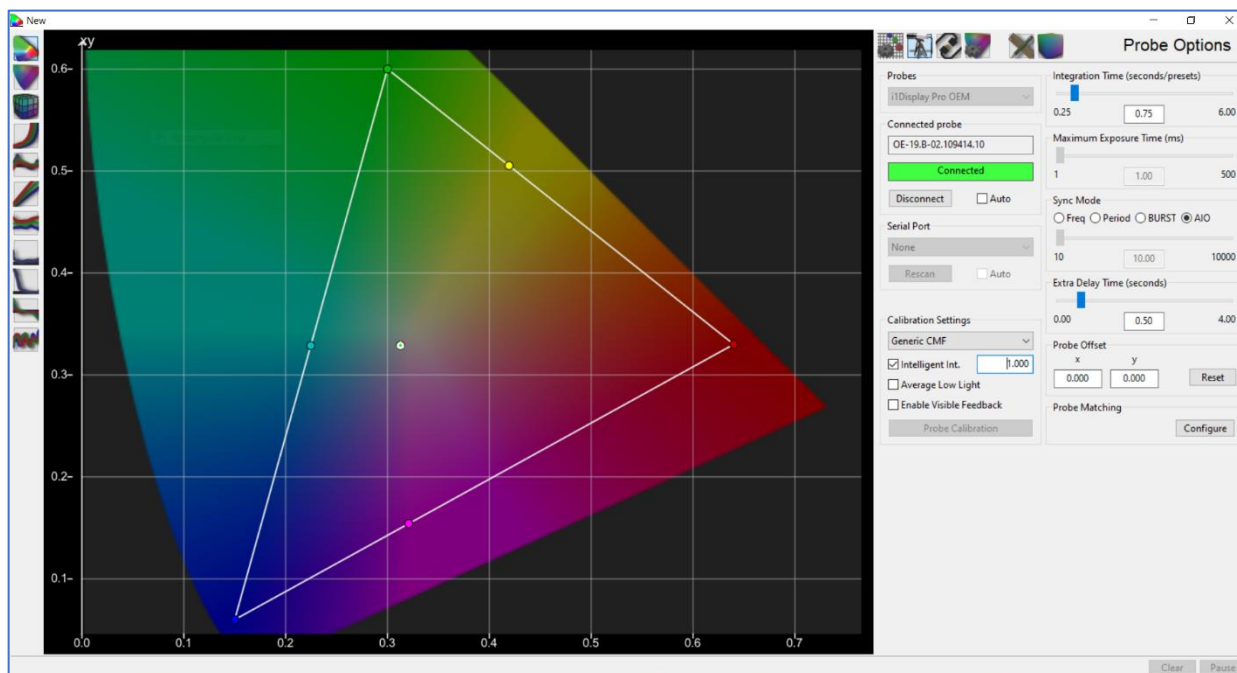
We will be targeting a peak luminance of 100 nits with 100% white, however, at this point there is no real need to set this here. Instead, place a check within the “Auto” box for Profile Luma. This will assist in setting the OLED Light setting by using the Manual Measure tab.

WRGB OLED displays benefit from Stabilisation. Setting this to 0.35 seconds and settings the patch colour to Black will insert a black patch for 0.35 seconds in between every patch that is measured. Ensure it is enabled by placing a check within the box.

When using the PGenerator for patch generation, we will be sending a Full Range 0-255 RGB signal. In fact, when using any patch generator, Full Range RGB is preferred when profiling for 3D LUT Generation as it allows for Whiter than White, up to 109% profiling. With Black Level Low set on the LG (*Limited on 2021 LG models*), reference 0% Black will be 16 and reference 100% White will be 235

At this point, there is no need to set a Pre-Roll, but remember this setting as we need to return to it shortly.

Switch to the Probe Options tab:



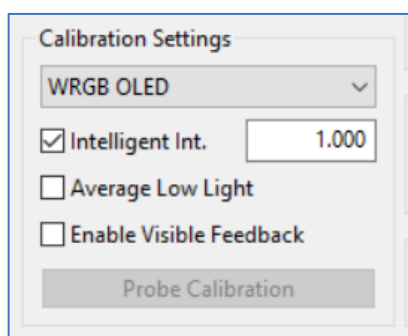
This is where we connect our measurement probe. For this guide, I have used the i1 Display Pro OEM (Rev.B). Select your probe from the available list and click Connect.

Note: The i1 Display Pro Retail is not officially supported, but it can be unlocked so that it correctly operates in ColourSpace CMS. Please Contact Ted Aspiotis via his [Contact Us form](#)

The recommended settings for the i1 Display Pro are as follows

- Integration Time 0.75 *Advanced users may perform tests in order to reduce this*
- Sync Mode AIO *Rev. B only*
- Extra Delay 0.50 *Recommended when using 24Hz signal on LG OLED*
- Intelligent Int 1.00 *Intelligently adjusts Integration time for low light measurements*

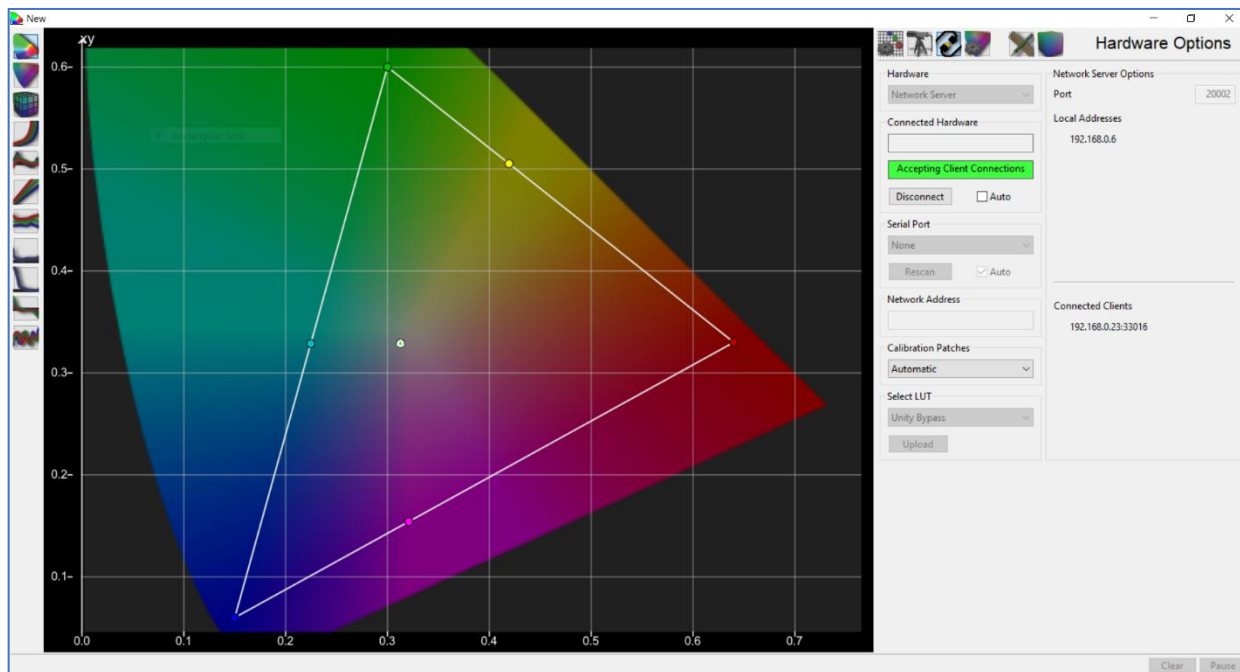
For users without a Spectroradiometer/photometer, it is advised to select the **WRGB OLED EDR** for use with LG OLED



The WRGB OLED pre-set EDR that is provided within ColourSpace is suitable for use on 2018, 2019 and 2020 LGD OLED panels. It was created using a professional FSI WRGB OLED monitor that uses the same underlying panel technology as all general consumer WRGB OLED displays from those model years.

For users *with* a Spectroradiometer/photometer who wish to “Probe Match” do not select a meter mode pre-set, select **Generic CMF**. You will also need to “Configure” the readings of each meter, but first we will need to connect the patch generator.

Switch to the Hardware Options tab:



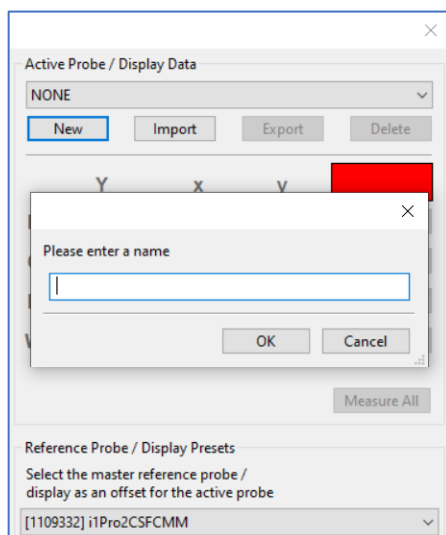
We will be using the Raspberry Pi PGenerator for this session which connects to ColourSpace using the “Network Server”, so select this and click Connect.

Set “Calibration Patches” to “Automatic”

Depending on your Test Pattern Generator, different options will appear to the right side of this tab.

For users who wish to Probe Match to a Spectro, return to the Probe Options Tab:

Connect the first probe, e.g. the Spectro. Click “Configure”. Select “None” and click “New”



When profiling a colorimeter to a reference (Spectroradiometer/photometer), it makes no difference as to which order you connect your probes to take the readings to create the correction. However, for simplicity, it is advised that you first connect your reference meter, take the required readings, and then switch to your target meter for correction (colorimeter). This is because you will be using the target meter for the rest of the process, and it prevents the extra step of switching probe connection again.

Never use a meter pre-set when creating a Probe Match in ColourSpace!!

For the i1 Display Pro, select Generic CMF

Enter a memorable name, such as the name of the Spectro with the date. This will make it easier to reference to at a later date. Once a name has been assigned, click “Measure All”.

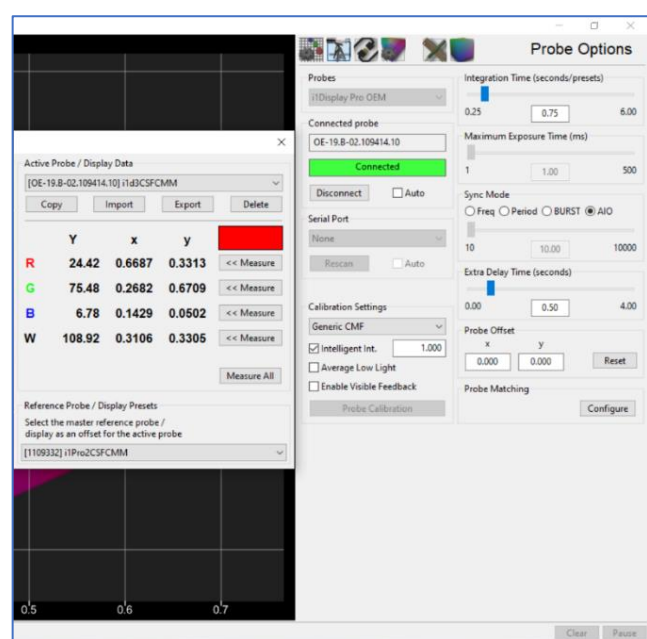
A series of Red, Green, Blue and White patches will cycle (3 of each). They will be measured and the average of each colour recorded.

When complete, disconnect the first probe and connect the second, e.g. The i1 Display Pro (there is no need to close the Configure window).

When Profiling to a Spectro, you should **not** use a meter mode pre-set. For the i1 Display Pro, select **Generic CMF**

Repeat the process as above, and when complete ensure that the Spectro file is selected in the “Reference Probe” selection box, and the i1 Display Pro file is selected in the “Active Probe” selection box.

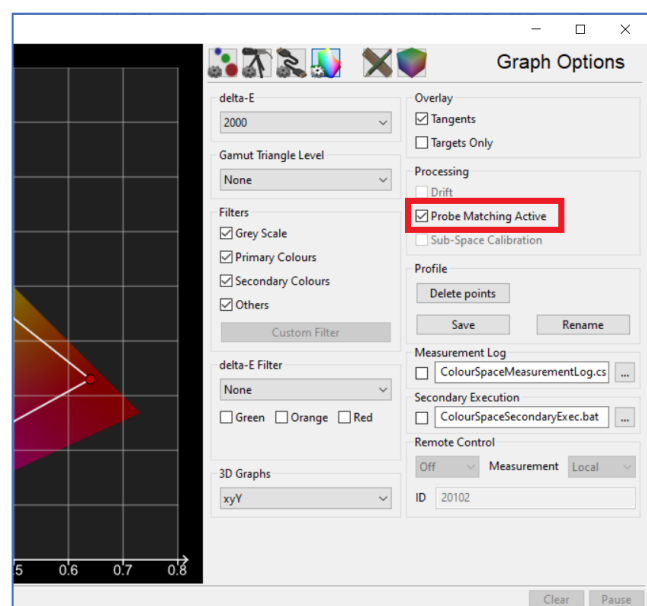
If done correctly, it should look something like this



Ensure that your target (colorimeter) file is set within the top selection box. The readings shown below are those taken by the target meter during configuration.

Finally, ensure that the reference (Spectro) file is selected in the bottom selection box. The readings from this file will not be visible.

The meter correction is applied “in software” when Probe Match Active has been applied within the Graph Options tab (See below)

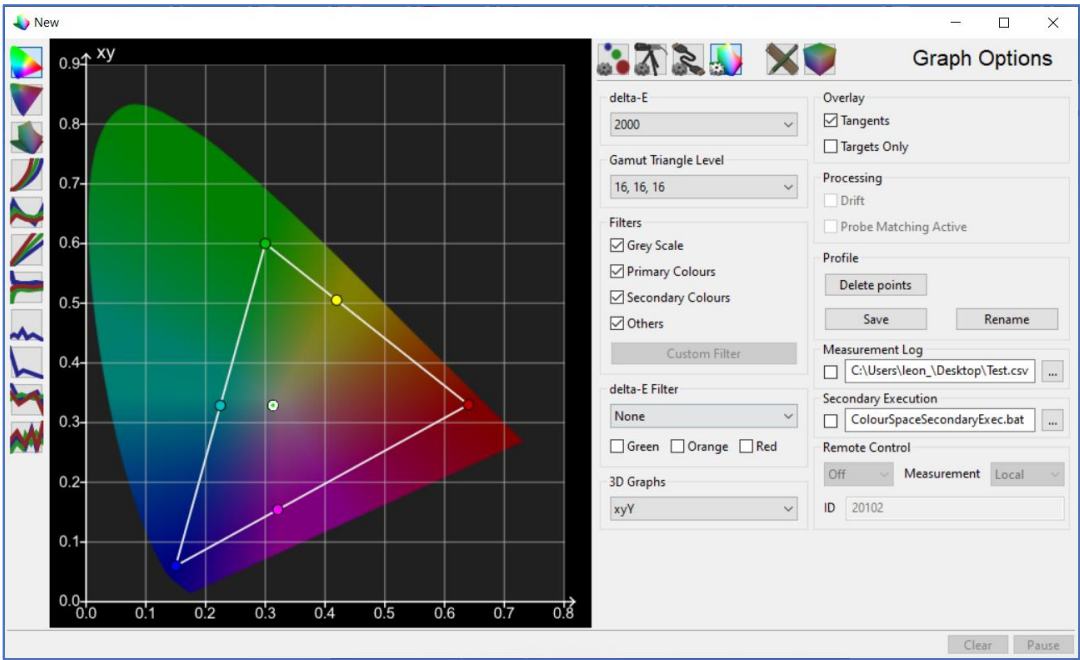


After Probe Matching has been performed, switch to the Graph Options tab and place a check mark into Probe Matching Active

Pre-Calibration – RGB Separation Check:

Switch to the Graph Options tab:

There is little to do here right now, but it is worth taking a look before we begin.



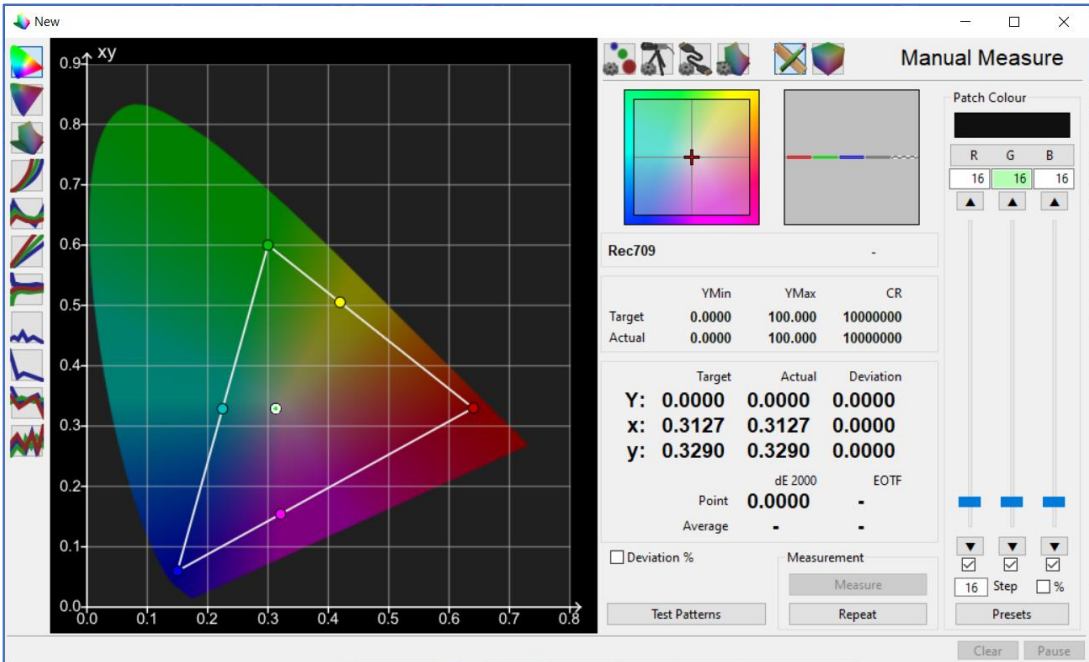
You can change the delta-E formula that is reported during measurements. The most common choice delta-E 2000

For users that have “Probe Matched” to a Spectro, you have a choice on whether to enable “Probe Matching Active” now, or after Profiling is complete. However, if you wish to take some measurements prior to full profiling it is advised to enabled here now. It is a personal choice, but for inexperienced users, it may be safer to leave it enabled at all times.

For users that have not “Probe Matched” to a Spectro, this function has no purpose, and will not be selectable.

Switch to the Manual Measure tab:

The Manual Measure tab is ideal for setting the OLED Light prior to calibration or during Service Menu White Balance pre-calibration.

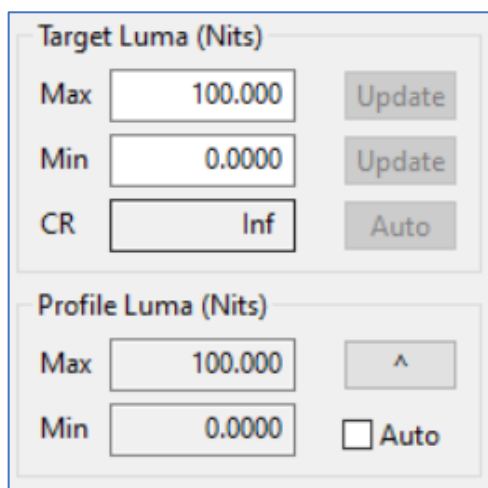


Placing a check below each of the RGB sliders will “link” them together. Set them all to the value of 255.

With the LG OLED set to **Black Level – Low (Limited on 2021 model LG’s)**, it will expect 16.16.16. to be 0% Black and 235.235.235. to be 100% White.

With Patch Scale set at **Extended** in the ColourSpace Settings tab, the RGB sliders will not be able to go below 16.16.16. 235.235.235 will still equal 100% White, and **255.255.255 will equal 109% White.**

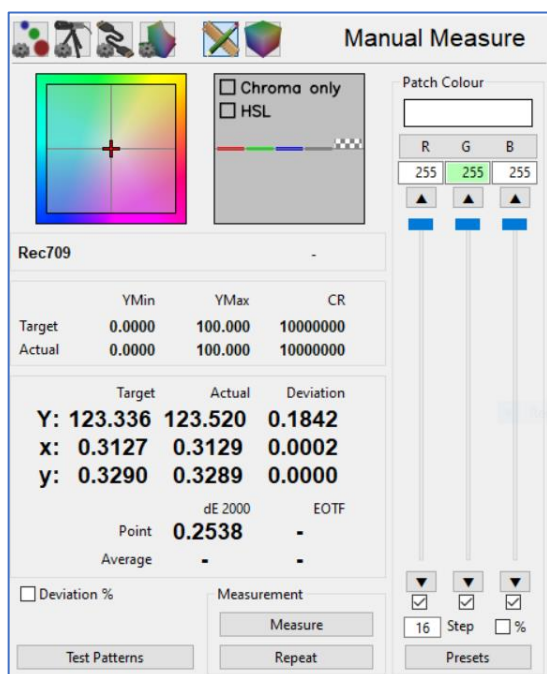
Tip for setting peak luma:



The screenshot shows the 'Target Luma (Nits)' and 'Profile Luma (Nits)' sections. In the 'Target Luma' section, 'Max' is set to 100.000, 'Min' is 0.0000, and 'CR' is 'Inf'. In the 'Profile Luma' section, 'Max' is 100.000, 'Min' is 0.0000, and there is an 'Auto' checkbox.

The Target Luma and Profile Luma options in the Settings tab are based on the “measured values” of 100% white (235.235.235).

You can manually enter your target 100% value into the Target Luma (e.g. 100 nits) which will automatically calculate your 109% target value based on the target colour space and gamma value.



The screenshot shows the 'Manual Measure' tab with a color checker chart and various measurement data. The 'Patch Colour' section shows R, G, and B values all set to 255. The 'Rec709' section shows YMin, YMax, and CR values. The 'Target' and 'Actual' values for Y, x, and y are displayed, along with the deviation. The 'dE 2000' and 'EOTF' values are also shown.

When switching back to the Manual Measure tab, when you set all RGB triplet sliders to 255, the 109% target will be calculated based on the target colour space and gamma (e.g. Rec709 2.4 Gamma).

This allows for an accurate 109% white peak target to be set based on personal preference for your eventual 100% target output.

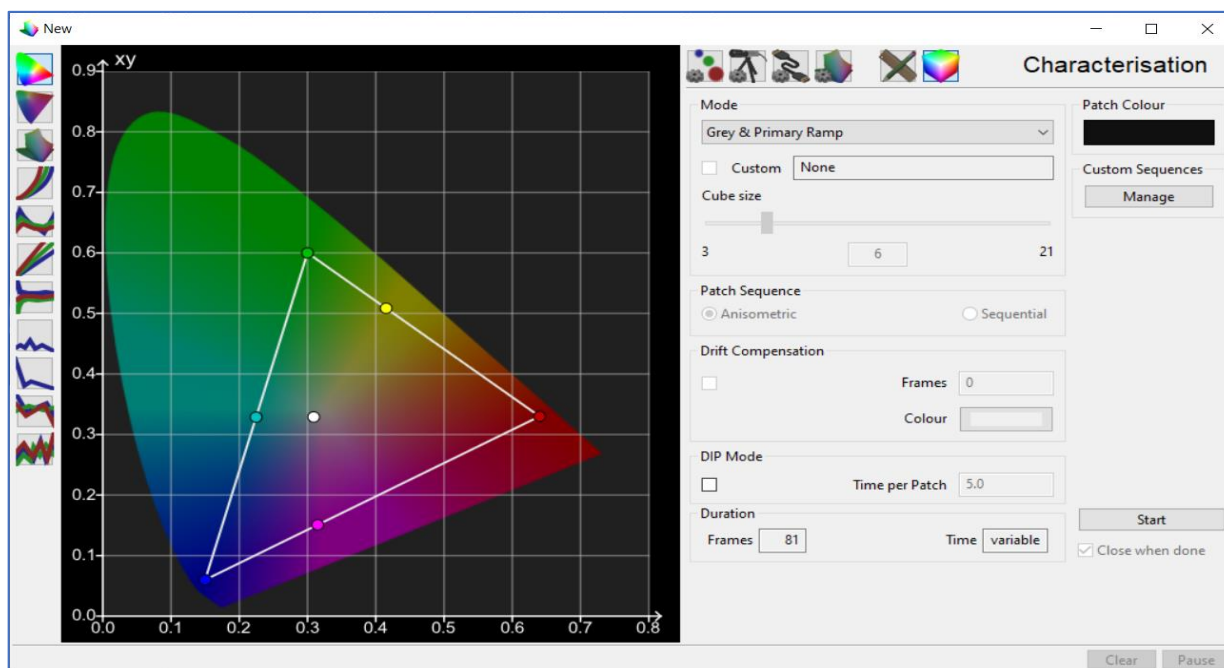
109% white is a required target for LG LUT calibrations due to how the internal LUT works.

There is also the option to load a custom created CSV patch sequence for some Pre-Calibration measurements, giving the opportunity to see how the native panel performs. It is not required for this guide as we will be using a built-in patch sequence to run a Pre-Calibration sweep, but the option is there for more advanced users should they wish to use it.

Use an RGB Triplet value of 255.255.255 to set your OLED Light value to output approximately 123-124 nits

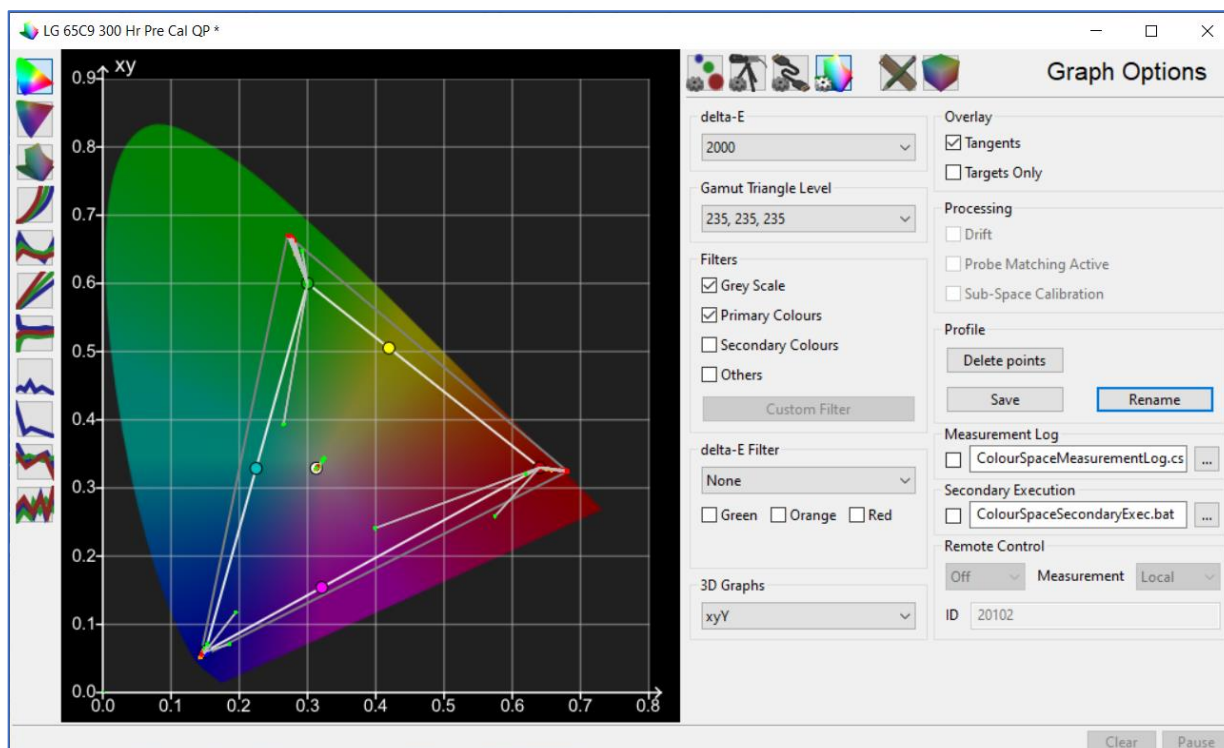
Switch to the Characterisation tab:

Before full profiling, we need to take a quick Pre-Calibration sweep of the Greyscale and Primary colours. This will help show whether the Pre-Calibration set-up and adjustments have been correctly performed.

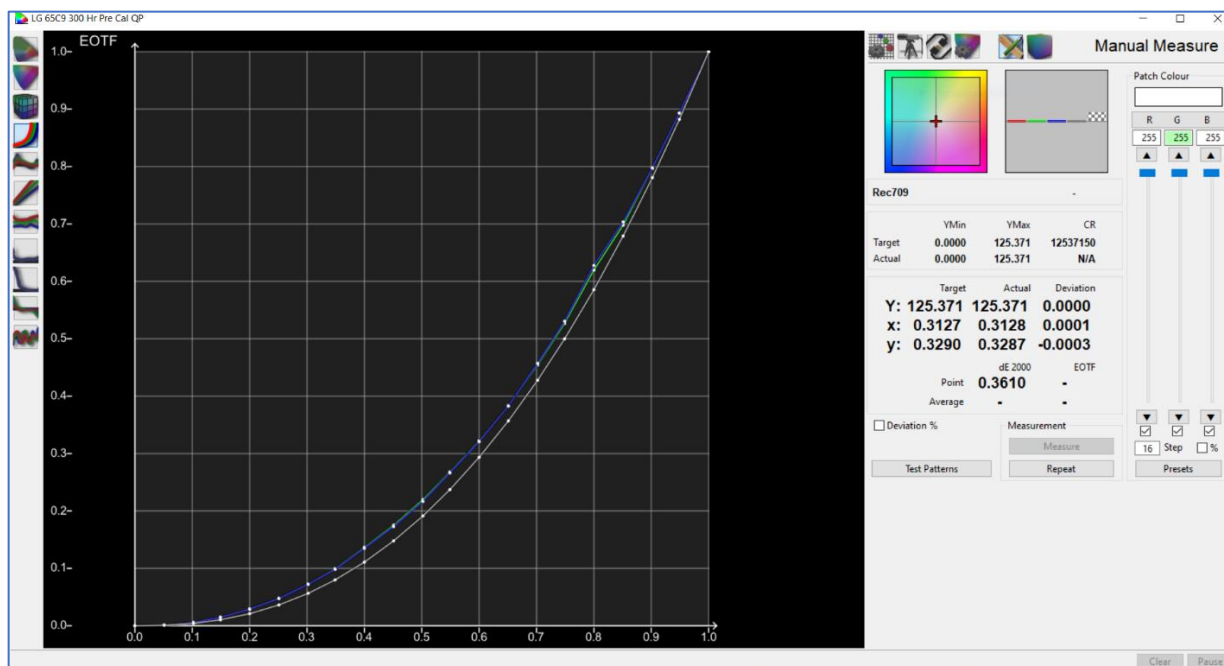


In the “Mode” selection box, select “Grey & Primary Only” and click “Start”. This will record 21-point measurements of the Greyscale and 20-point of the three primary colours.

Once complete, the CIE graph should be plotting the primary colours outside of the Rec709 gamut triangle which indicates the display has been measured in its native gamut. If this is not the case, then a step has been missed so it will be required to return and ensure each step has been performed correctly.

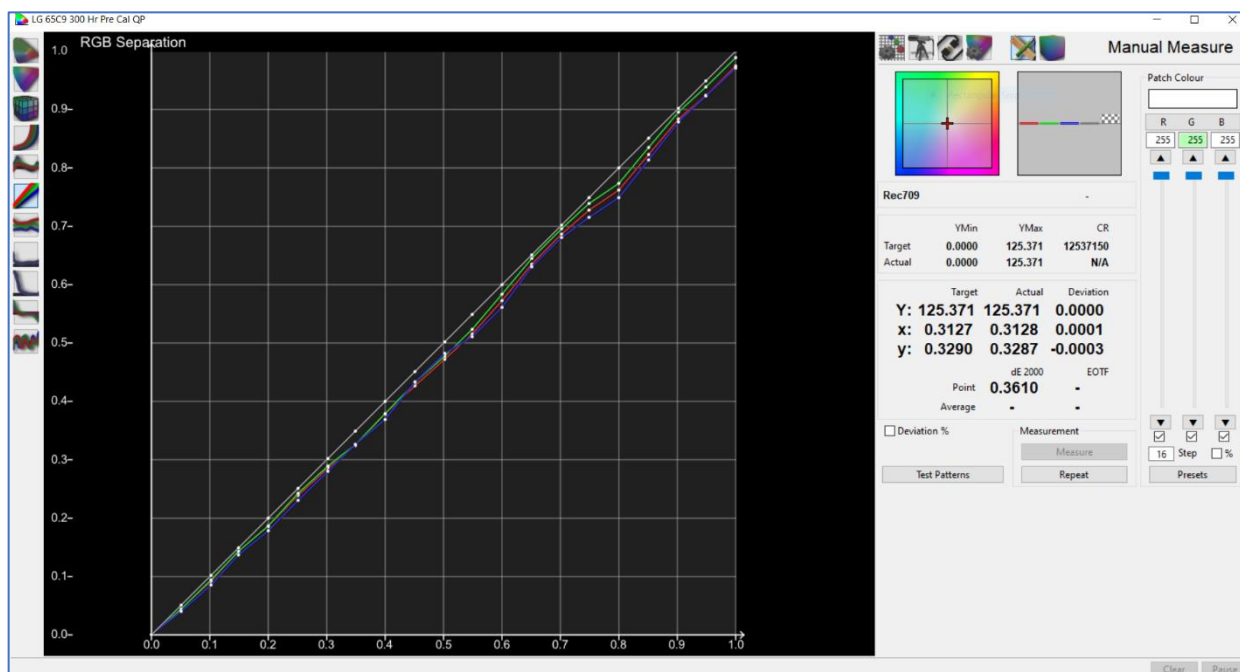


Using the icons to the left of the CIE Graph, switch between graph types, beginning with EOTF



The solid white line is the target EOTF, which in this case is 2.4 Gamma. As you see here, the measurement is tracking above the white line (brighter) meaning a lower gamma has been recorded. Remember we have measured the panel native 2.2 Gamma, so this is expected and is correct.

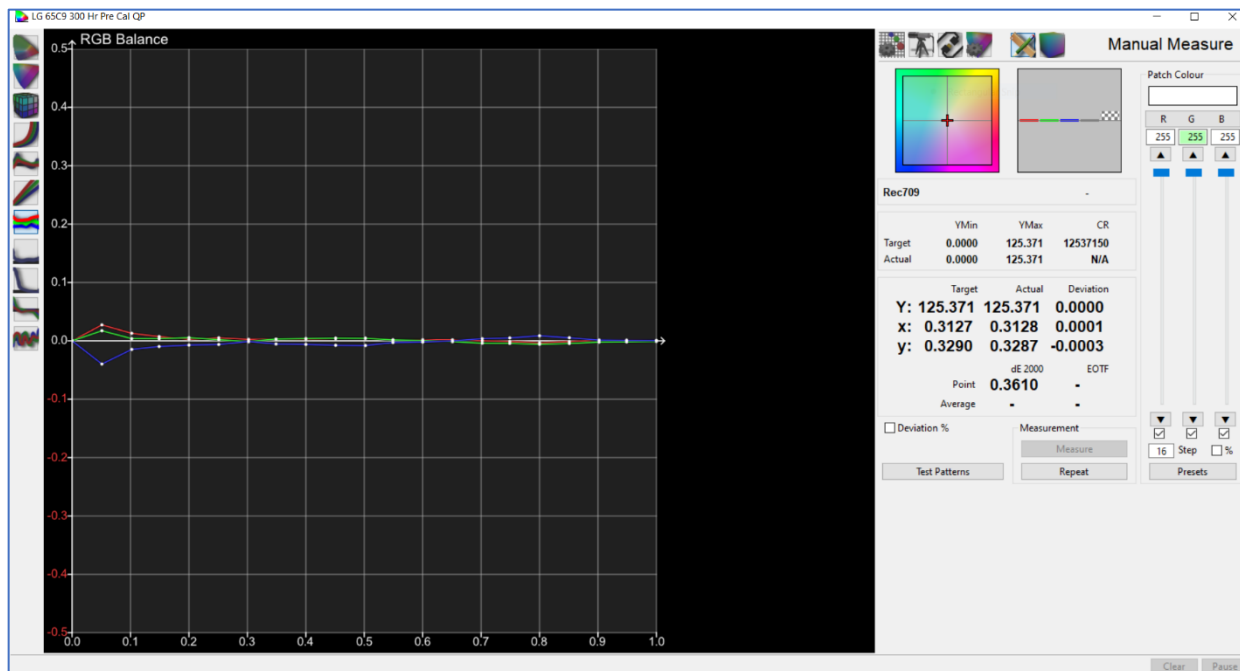
Switch to the RGB Separation Graph



It is not going to be perfect; the display is in an “uncalibrated state”, but there are no plateaus which shows no colour clipping. If there are significant deviations in this graph, or you notice the RGB lines plateau towards the top end, then it will be required to “fix” RGB Separation. You can find information on how to do this by following the link below and reading the subsequent posts.

[RGB Separation – Examples of poor RGB Separation and methods used to fix](#)

Switch to the RGB Balance Graph



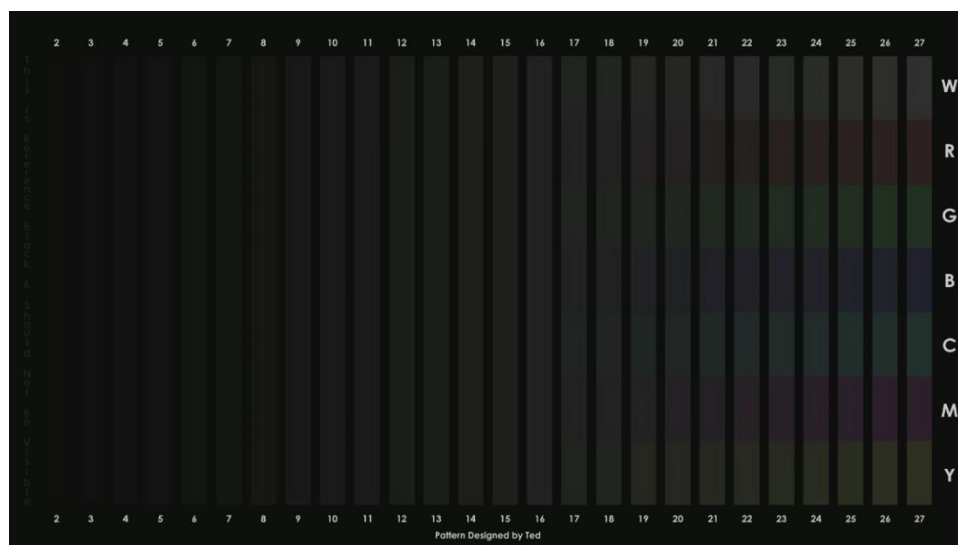
From one simple adjustment in the Service Menu to correct White Balance at peak white, it has balanced the entire Greyscale rather well.

With this one simple Quick Profile of Greyscale and Primary Colours, we can see that the display is ready for full profiling and LUT generation.

But first!

It is advised that you check one final time your Brightness setting is correct by using a verified accurate Brightness or Pluge pattern.

Ted's LightSpace Calibration Disk, Advanced Brightness Flashing bars is highly recommended for this purpose



Please note that the image to the left has been captured in Full Range, which shows all the bars below Video Range reference black.

*With your display correctly configured, you should **not** see anything up to, and including 16*

[Ted's patterns are available as USB mountable media files and can be purchased by visiting his website.](#)

[They are also available directly integrated within PGenerator](#)

Brightness should be checked from up close to the screen, not from your normal seating/viewing position. The bars up to and including 16 should **not be visible**, bar 17 **should be barely visible** with the bars above 17 rising in luminance. The visible bars will be flashing when using described the pattern above.

Display Profiling

Pre-Roll:

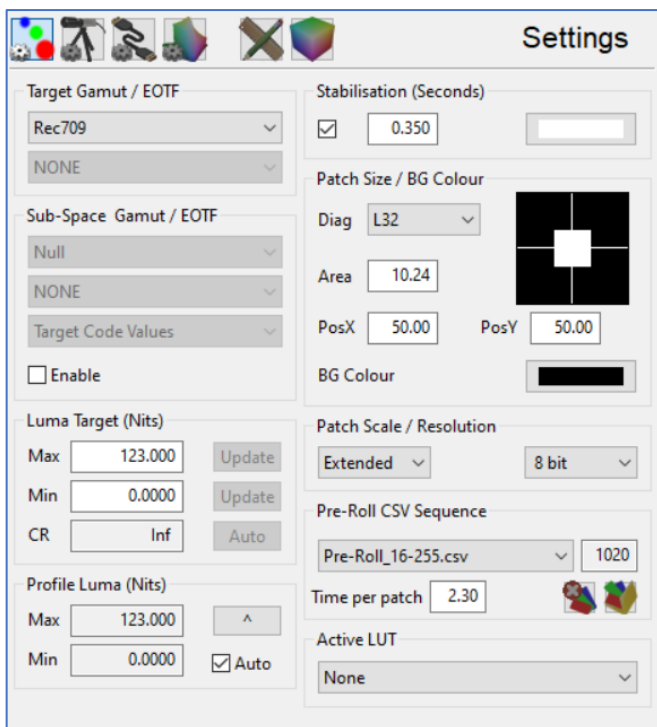
The following sections of this guide will work on the assumption that you have been provided with “Special” patch sequences that have been specially designed to produce consistently excellent results on LG OLED displays

WRGB OLED displays are highly unstable by nature. Prolonged profiling with random patches can cause the panel to heat up and cool, meaning measurements can drift wildly. This is more evident in very high luminance ranges, such as HDR. But even in SDR, the panel can drift significantly. To counter this, a Pre-Roll patch sequence is desired.

The Pre-Roll patch sequence will run for a pre-determined time to bring the panel to a stable temperature, at which point the main profiling patch sequence will begin.

You will have been provided with three sets of CSV patch sequences. These have been named in a logical manner for convenience (Pre-Roll, Profiling, Verification).

Remember when we opened the main Profiling window for the first time, I said to remember the “Pre-Roll” settings as we would return to it later. Let’s go back there now.



Whilst loading the Pre-Roll.csv, take the time to once again check that the following settings are correctly configured.

- Target Gamut Rec709
- Stabilisation 0.350
- Patch Size L32
- Patch Scale Extended
- Resolution 8 bit
- Pre-Roll Time 2.30

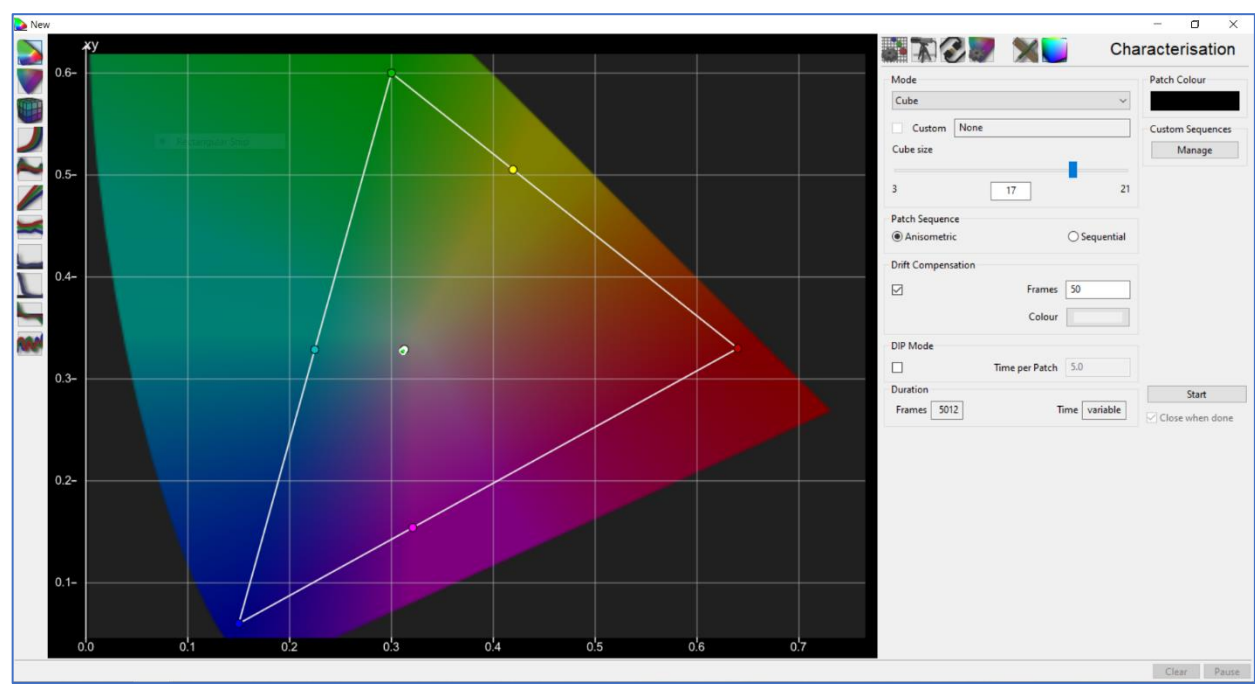
Click the folder icon to “Load” the Pre-Roll CSV patch sequence. A file explorer window will open to browse for the file’s location. Select it to load.

For “Time per patch”, the recommended value is 2.30 seconds. It is advised to use this recommended setting for your first profiling session.

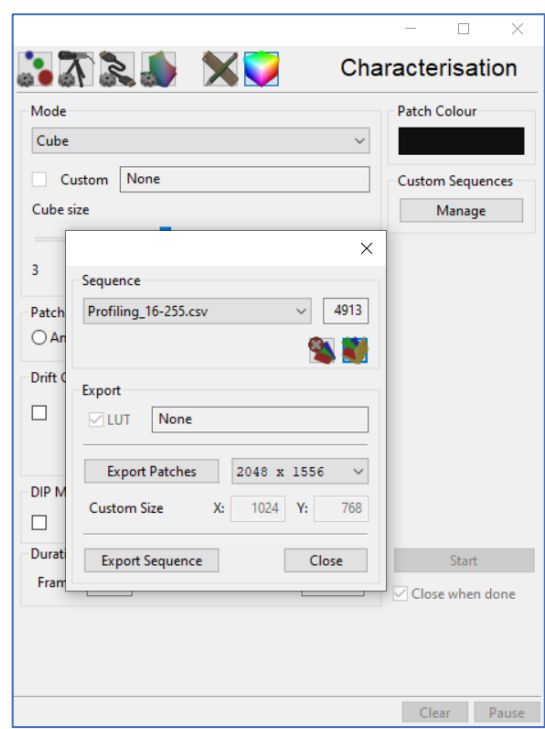
Through experimentation in your specific environment with differing peak luminance values it can be adjusted higher or lower. As more experienced is gained then you can experiment with various methods such as running different sequences with Drift Compensation enabled and with close monitoring of the drift chart you may find a lower (faster) value. For beginners on their first attempt, I highly recommend staying with the 2.30 value to for now.

Characterisation:

With the Pre-Roll loaded, return to the Characterisation tab



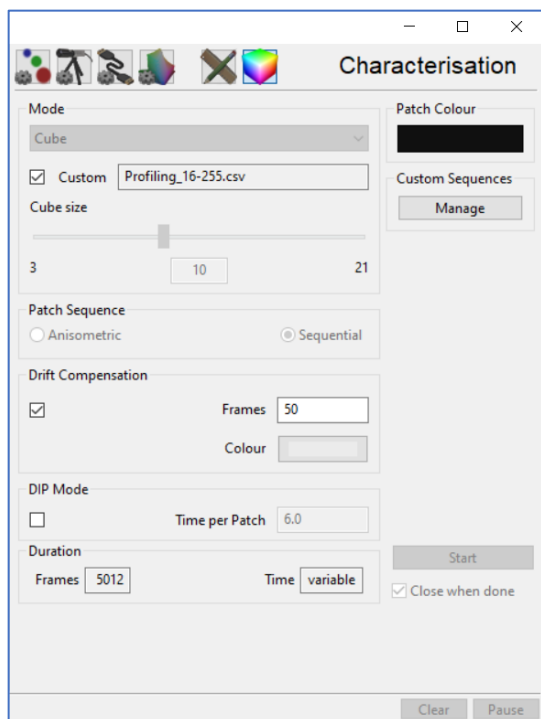
You can either select a built in “Cube” sequence, or if you have been provided with the “Special” patch sets, it can be loaded by clicking “Manage”



When you load the Profiling.csv sequence, the number of points contained within will appear. The sequence provided with these guides contain 4913 points

Once again, click the folder icon to load the “Profiling” CSV sequence. You will see the number of points contained within the set appear to the right. Once loaded, click “Close”

Place a check in “Custom” within the “Mode” area



Note that when you enable “Custom” the Mode selection becomes greyed out.

When you enable “Drift Compensation” the value entered here will add additional frames to the total profiling sequence. This is normal.

Finally, enter a “Drift Compensation” value of 50.

Drift Compensation inserts a patch (default 94% Grey) into the sequence at a pre-determined interval (in this case, after every 50 patches measured) which is recorded throughout the profiling session. The variation in luminance and chromaticity of this drift patch can be viewed in the Drift Plot graph during, and after profiling has completed.

During LUT generation, the drift patch values can be analysed and accounted for if any panel drift has occurred during the profile. Drift does not have to be used, but through extensive testing, it has been found to provide the best results on WRGB OLED displays so it is advised to use the suggested setting.

With all the Pre-Calibration adjustments made and checked, the Quick Profile performed and checked, and all settings entered and of course checked, click “Start” to begin the Pre-Roll which will automatically be followed by the main Profile.

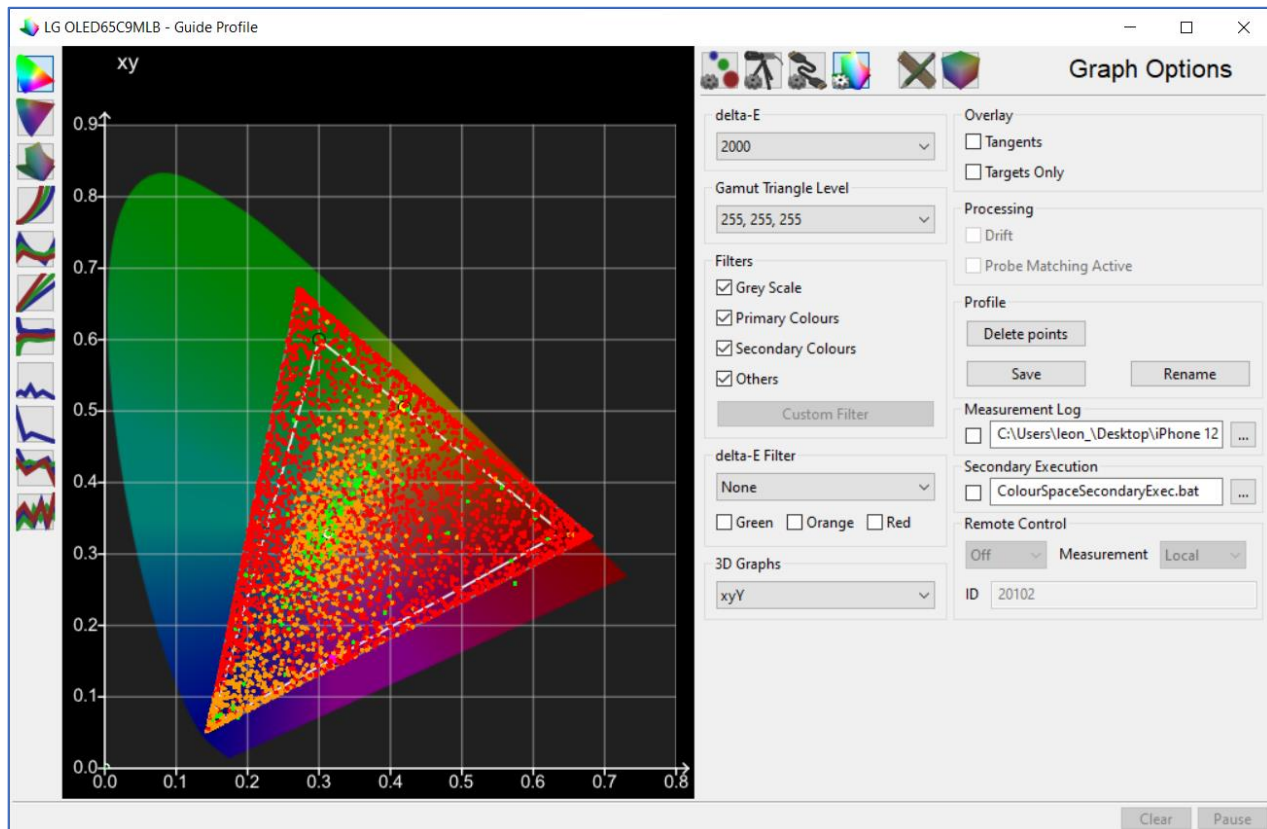
Now just wait.....

LUT Generation

LUT Generation and upload:

Once the display profiling has completed, it should be saved in the Graph Options tab.

Post Profiling checks:

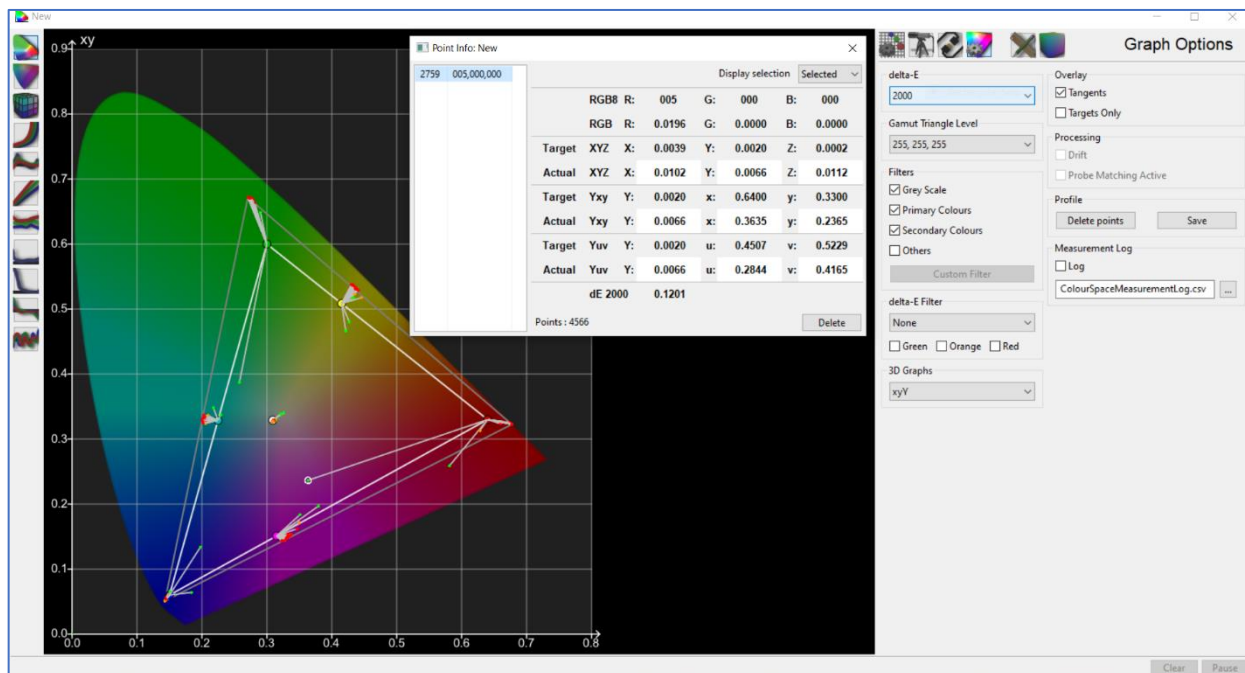


Click “Rename” to name the measured profile data, followed by “Save” to add it to your library. You can now take the opportunity to examine the graphs if you wish.

Notice how the measured points are mainly Red (above delta-E 2.3) and outside the target Rec709 gamut triangle. This is as expected as the native panel has been measured. In this tab, you can enable “Tangents” in the top right Overlay section. This will apply a tangent line to each individual measured point directed to where it actually should be if it were calibrated to reference.

As you become more familiar with the graphs in ColourSpace, you will begin to see how they give an unrivalled representation of your display’s capabilities. To learn more about the graphs and what they report, please visit the [Light Illusion website links provided earlier in the guide.](#)

With the use of “Filters”, you can enable or disable various aspects of the measured points. With “Tangents” enabled, I could easily spot a very large tangent toward one of the primary colours, so by removing the check from “Others”, we can examine this more closely.



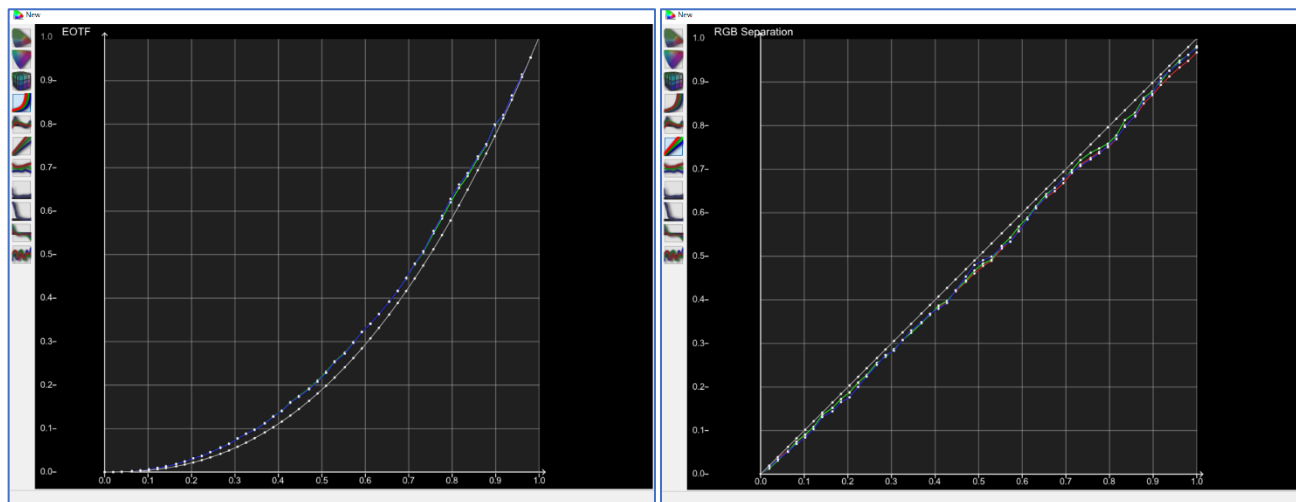
By selecting (double click) that individual point we can see detailed information on it in the Point Info window. Using this information, it is clear to see that the larger than usual “error” is due to the i1 Display Pro’s limited capability in low light readings. The recorded data for this particular point is 0.0066 nits, far below the threshold of 0.1 nits that X-Rite certify the accuracy of the probe at.

You have some options at this point.

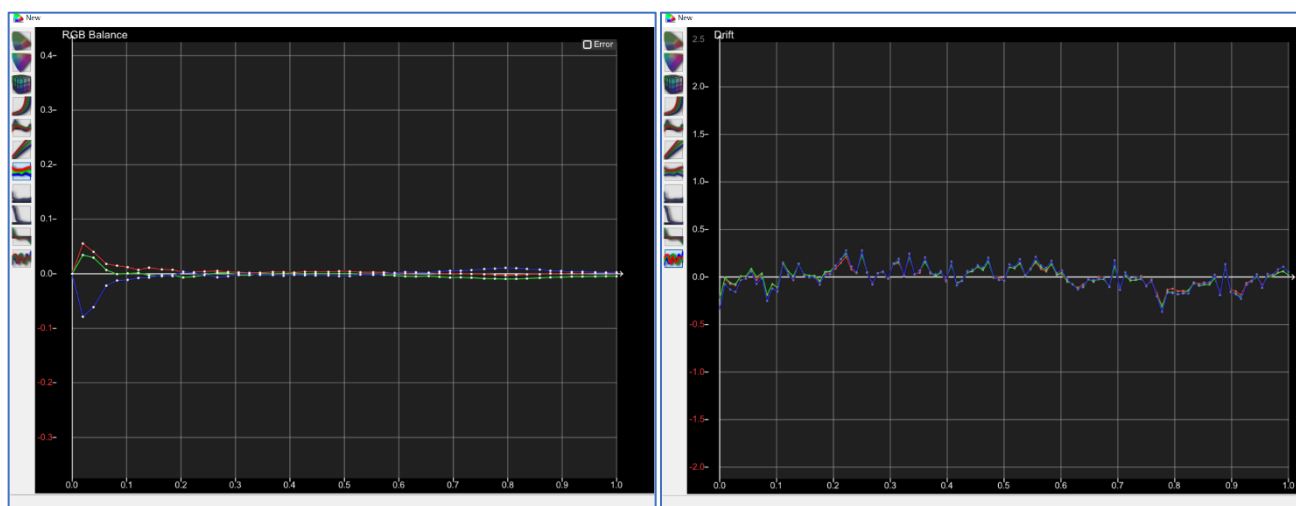
- You can attempt to “Re-Measure” the point by going to the Manual Measure tab. The selected point will already be pre-selected here too, just click the “Measure” button. *(this function will be moved directly to the Point Info window in the future)*
- You could “Delete” it altogether and allow the LUT Generation to correct for it during the correction calculation.
- You could leave it. The Colour Engine in ColourSpace CMS can intelligently disregard “outlier” measurements such as these during the correction calculation.
- You can also save a profile using each of the above options and generate a LUT with each of those profiles to compare the difference, if any.

With readings of such low light output as above, it doesn’t really matter which of the options you choose as the i1 Display Pro has limited capability at such luminance levels. But this method can be used to pick out any potential “bad reading” during the profile prior to LUT Generation.

Check through the other graphs if you wish, if only to familiarise yourself with an approximation of how they should look after a successful profiling session.



As before the EOTF will track above the target gamma of 2.4 and RGB Separation does not plateau



RGB Balance is still fairly good throughout the entire Greyscale due to the Service Menu White adjustment, and the profile drift is minimal due to the combination of using the Pre-Roll and the Optimised Patch Sequence.

Please note: The Drift Plot as shown here is exceptional, but is not always achievable. Meter placement and environmental conditions can also affect drift, as well as the patch sequence used for both the Pre-Roll and the Profiling. Do not be concerned if you cannot achieve such a “flat” drift plot, a range of +/- 0.5 can still be considered very acceptable on WRGB OLED.

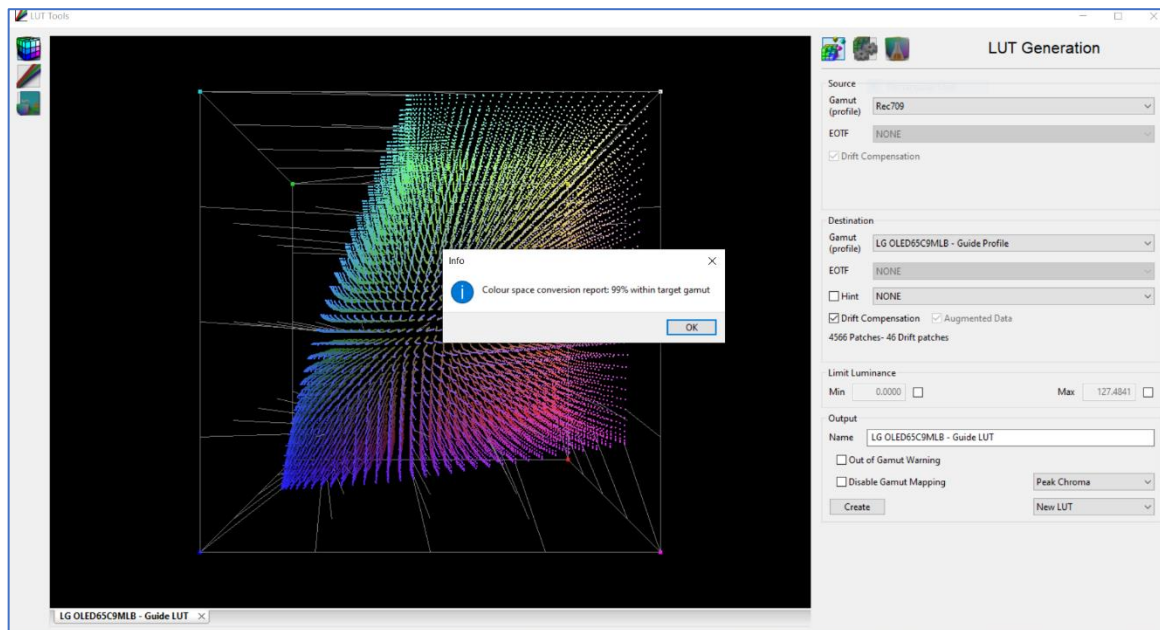
Generate LUT:

From the main ColourSpace CMS window, click **LUT Tools** to open the main LUT Generation window.



Depending on software licence level, there may also be various LUT Manipulation tools here, but in this case, we won't be using them as they are not required.

From the main LUT Tools window, set the correct settings for LUT Generation:



For Source, select the “Target Gamut”. In this case we are targeting a reference target of Rec709.

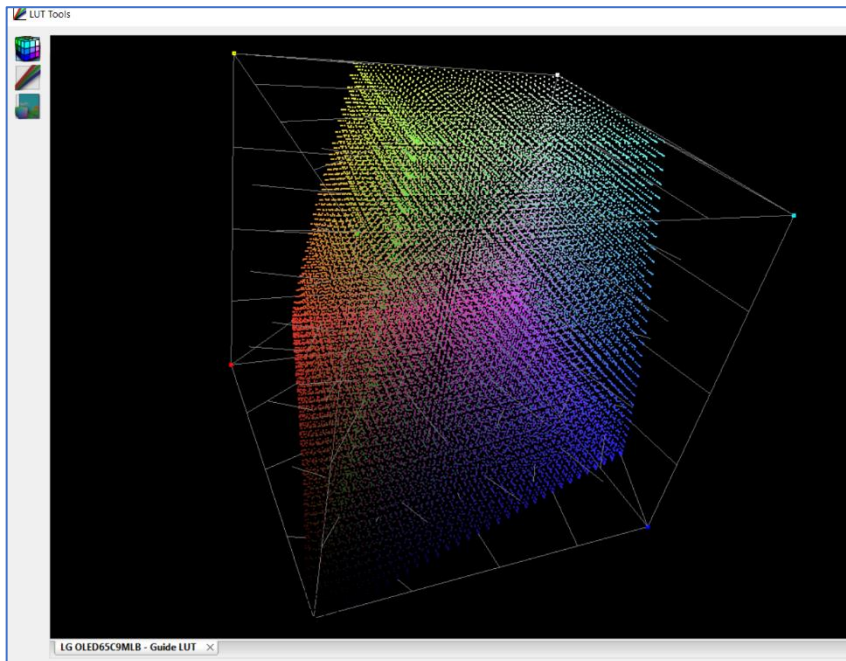
For Destination, select the profile you have just created. Be sure to enable “Drift Compensation” to allow the drift patches that were inserted to compensate for any potential drift during profiling.

For Output, give the LUT a name, ensure “Peak Chroma” is selected and click “Create”.

You of course have the option to generate multiple LUT's with drift Enabled/Disabled giving the chance to compare the two if you so choose.

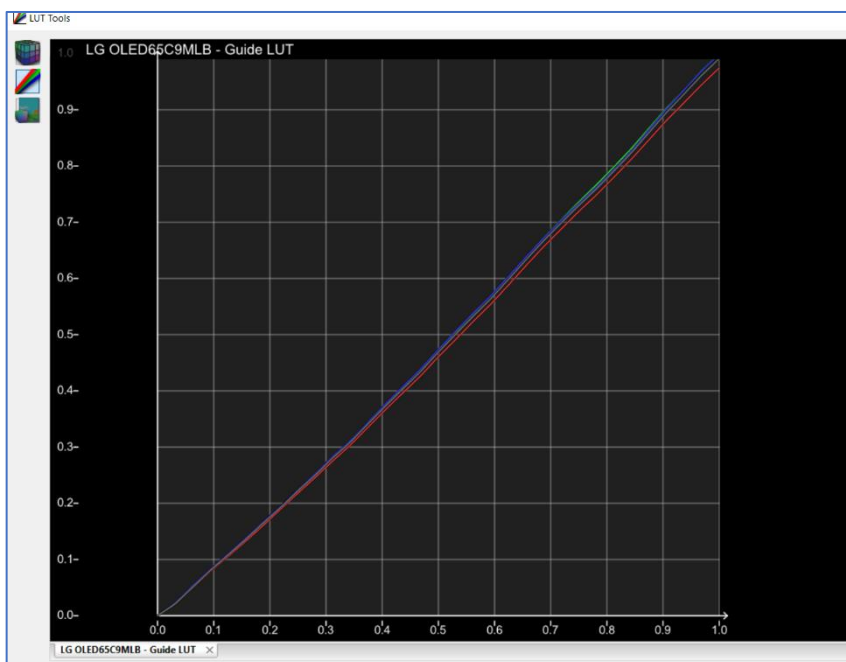
Using the three graph options on the left, examine the correction LUT:

Use the mouse to rotate and zoom the correction LUT. It should be smooth and linear



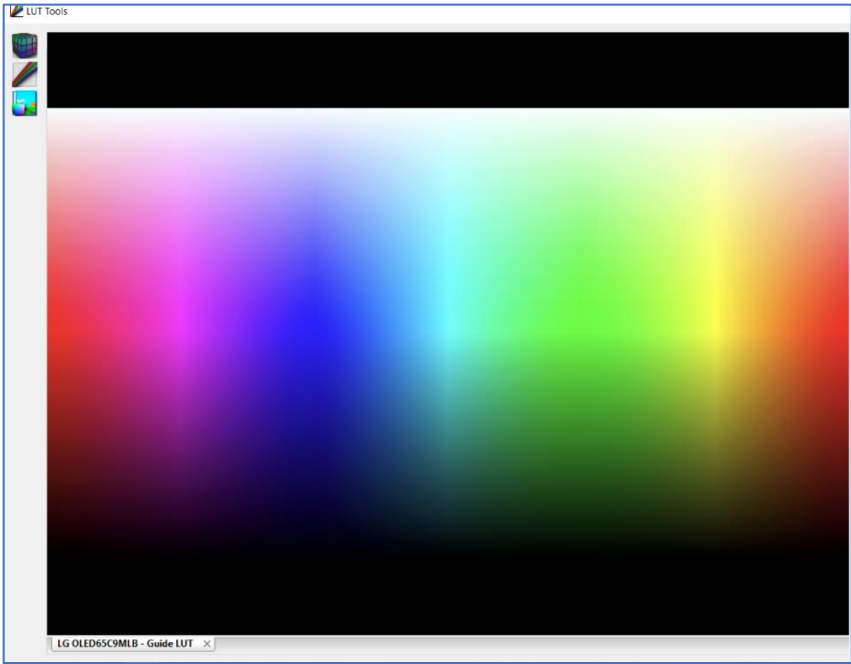
Note: The 3D Cube View should look similar to this. Imagine the “Outer Cube” as the native gamut of the display, with the inner “Compressed Colours” as the required correction to achieve the target gamut. For LG OLED displays, this equates to the native display reaching close to P3 gamut being reduced to a target Rec709

Use the 1D LUT View to see the required correction that is made to the Greyscale and Gamma



The 1D LUT View shows minimal RGB Balance correction thanks to the Pre-Calibration of White prior to profiling, however there has been some luminance adjustment to correct from the panel native gamma of 2.2, to the target gamma of 2.4

Use the LUT Image View to apply the correction LUT to a Colour Ramp image.



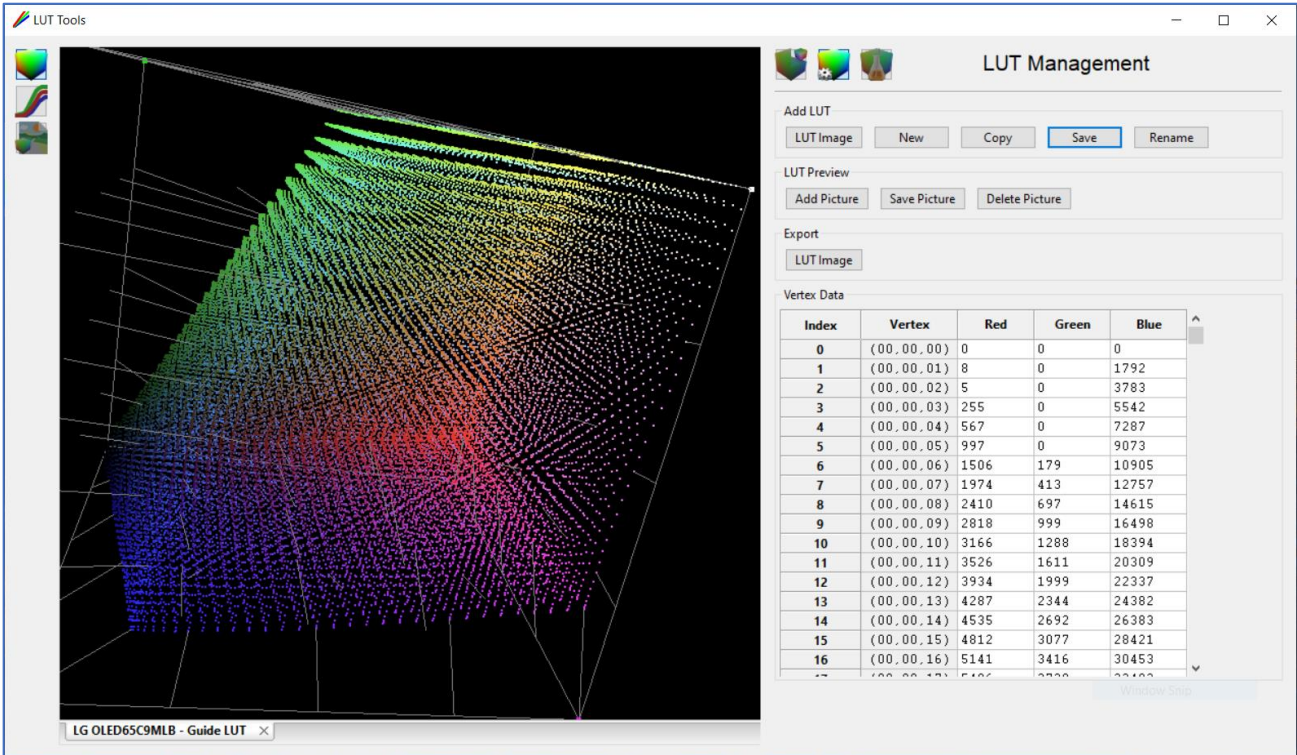
If you have your own reference images, you can load them via the LUT Management tab.

Select Add Picture, and browse to the location of your reference images.

This allows you to see how the correction LUT will affect the image

Hovering the mouse over the image will show an option to “Apply LUT”. Toggle this on and off and examine the image. A successful LUT will not introduce any negative effects to the image, such as banding or artefacts

Finally, navigate to the LUT Management tab to save the LUT to your library.



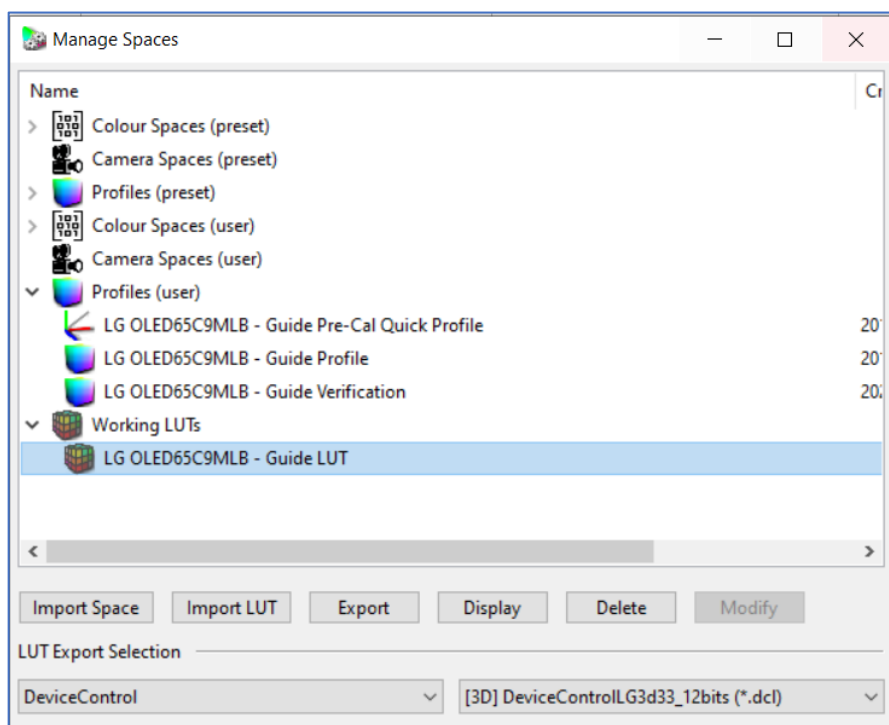
Export the LUT for Upload:

The correction LUT will be uploaded to the TV using Device Control and the correct Template for your model year TV.

From the main ColourSpace CMS window, select “Space Management”.



This will open a window containing (among other things) the profiles you have created.



The LUT will need to be Exported in the correct format.

Select “DeviceControl” followed by “[3D] DeviceControlLG3d33_12bits(*.dcl)”.

Please note, for “B-Series” LG OLED owners you must select “[3D] DeviceControlLG3d17_12bits(.dcl)”.*

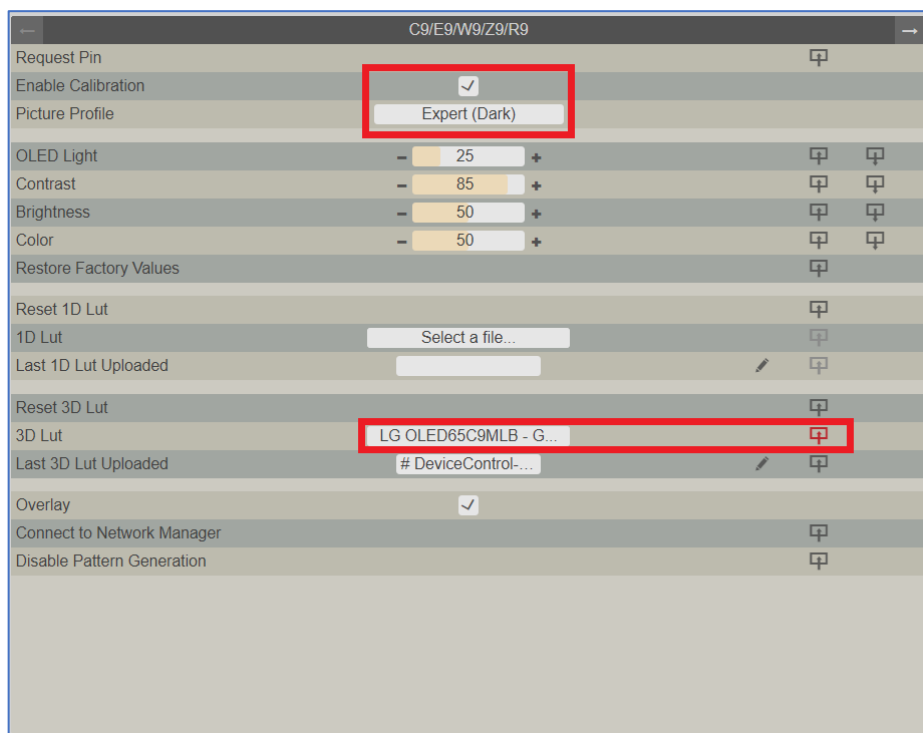
Highlight the generated correction under “Working LUTs” and click Export. Save the LUT in a memorable place (e.g. Desktop) for easier location when browsing for the LUT in Device Control Template.

Upload the LUT to the TV:

If you have left Device Control running and the browser window with the GUI/Template open or minimised, you will still be connected to the TV and there will be no need to reconnect.

However, if you have closed either of these or have somehow lost the connection during profiling, then you will need to relaunch Device Control and the web-browser, navigate to <http://127.0.0.1:81/> and reconnect to the TV as described at the beginning of this guide.

With Device Control Template open, click to “Enable Calibration”.



Please note that by default, each time that you “Enable Calibration”, the LG will always select the Cinema picture mode to begin with. Please ensure you correctly select the target picture mode that you made all your pre-calibration adjustments to and performed the main profile in.

With the correct picture mode selected, click on the entry field box beside 3D LUT and browse to the location of your exported LUT file with the resulting file explorer window that opens. When selected, click the command to upload the LUT to the TV. This should take just a few seconds.

Device Control is no longer needed, so you can now “Disable Calibration”, close the browser window and Device Control.

Verification

Verify the uploaded LUT:

Alternate calibration software solutions use rather small sequences consisting of just a few Greyscale points and some memory colours for verification. Unfortunately, this does not give a full volumetric representation of how good the calibration actually is. Measuring 10 Greyscale points and just over 20 colours cannot tell you the final accuracy of a volumetric 3D LUT calibration.

With ColourSpace it is generally advised to use a full volumetric validation patch sequence consisting of 1000 points. This can, and does, give a highly accurate verification of how good the outcome of the calibration is.

To do this, you can use the existing profiling window (if you have left it open) as beginning a new patch sequence will automatically delete the existing profile data.

If you have closed the existing profiling window, open a new one ensuring to set the same parameters for the options described previously, and connecting both the probe and patch generator in the same way.

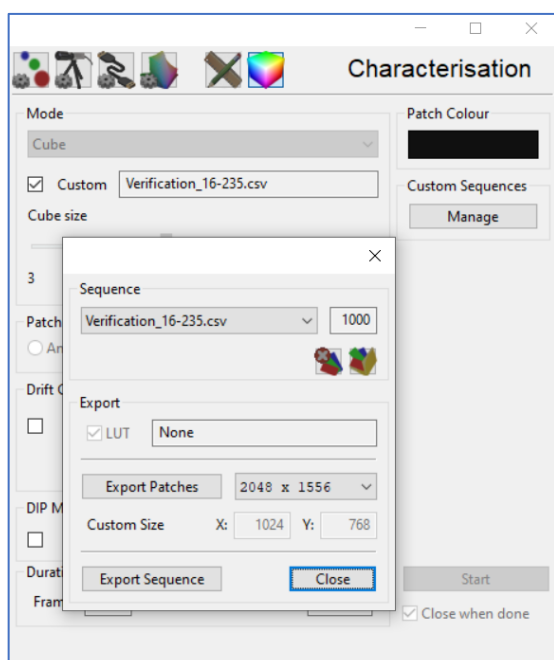
However, for verification purposes there are some slight differences.

- There is no need for a Pre-Roll so you can remove this to save time if you wish *Optional*
- There is no need for Drift Compensation so this can also be removed *Optional*
- Set Patch Scale to Legal *Settings tab*

Setting the Patch Scale to Legal in the main “Settings” tab will verify the calibration in the video legal range of 16-235 (0% to 100%). Verification above 100% is not required.

Switch to the Characterisation tab:

As before, if you have been provided with the three “Special” patch sequences, one of them should be the verification sequence. This should be once again loaded using “Custom Sequences”.



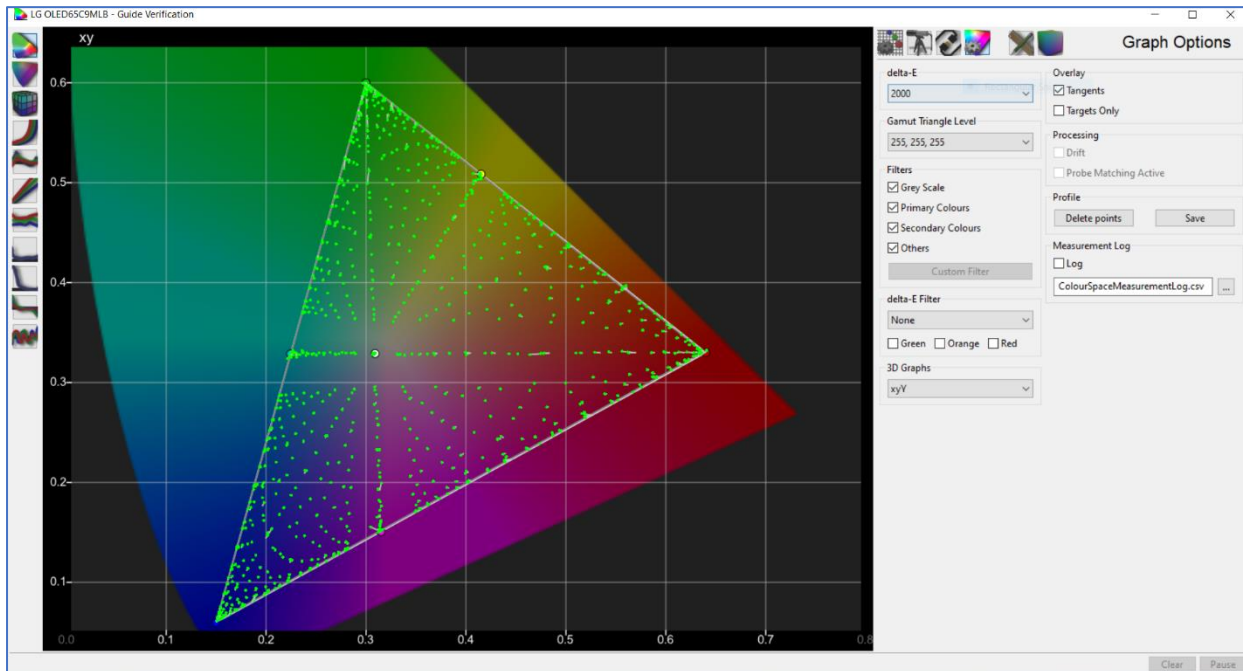
Close the Custom Sequence manager and click “Start” to perform the 1000-point verification.

Upon Completion, you will be presented with the outcome of your calibration.

View the outcome:

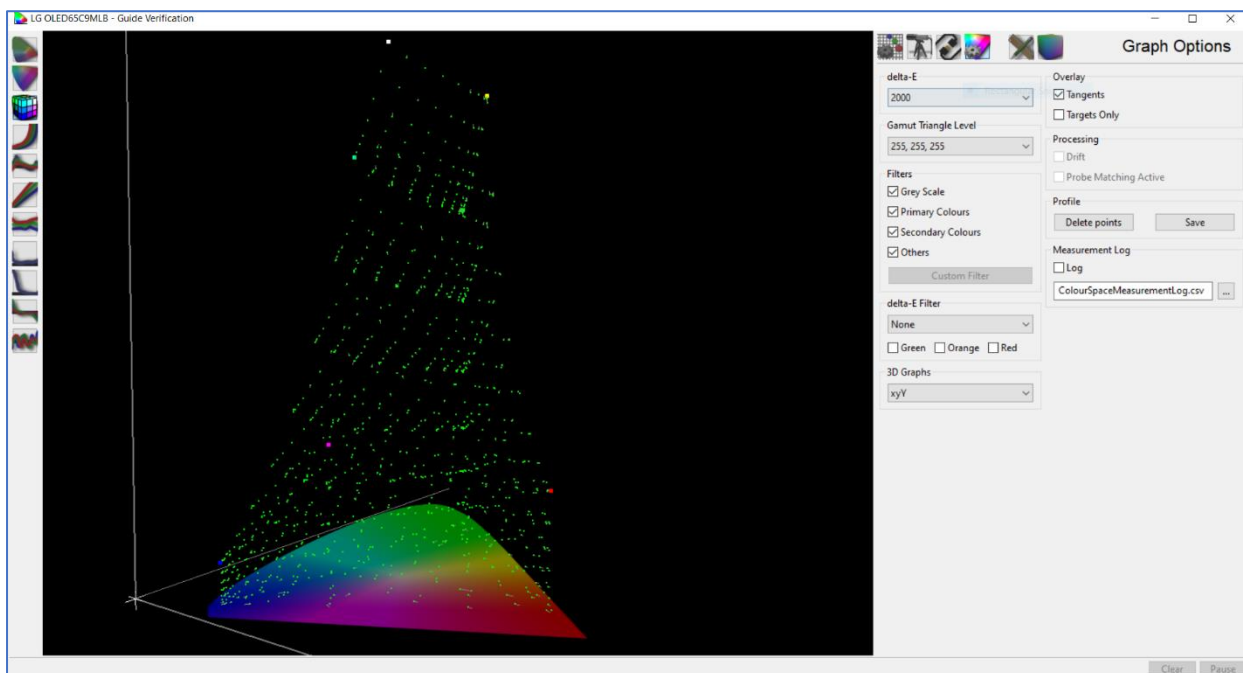
Switch through the various graphs and tabs to see full and comprehensive data on the verification profile.

Using the standard CIE 1931 xy Graph, it is already obvious that this particular calibration has produced excellent results



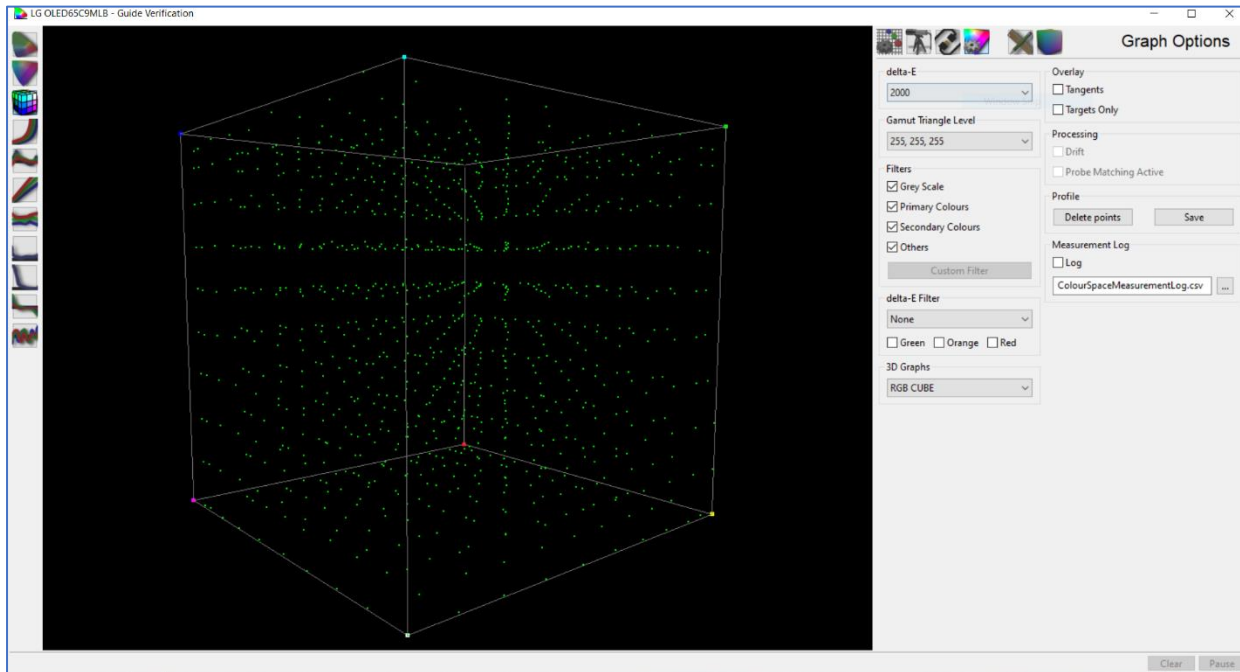
With all filters disabled, it is clear that all measured points have resulted in a delta-E of less than 1.0. These are phenomenal results and the resulting image could be considered reference quality.

Switching to the 3D Graph and selecting xyY in Graph Options will show the results volumetrically with luminance



Once again showing the accuracy of correction LUT.

One final example. Change the 3D Graph to RGB to be presented with a “normalised” cube view



With the generated LUT Cube Viewer, the outer cube was the native panel gamut, with points inside compressed to correct to Rec709. Here the outer cube is Rec709. As you can see, all points fit inside in a near perfect linear cube formation. This means reference accuracy.

Reports:

At the time of writing, Calibration “Reports” had only just been implemented within ColourSpace CMS. Reports are based on a template that users can create themselves, which may seem daunting and complicated for the average new user to create. [Light Illusion are offering advice and support to those wishing to create a template](#), and will offer generic sample templates for customers to download and import, but it is early days and the templates are currently limited.

At some point in the future, when the Reports function is fully implemented and finalised, a standardised template for LG OLED calibration and verification will be created and this may possibly be included in a future revision of this guides pack, or may just be made available to be downloaded from the Light Illusion forum thread linked above.

Thanks to AVSForum user [Anger.miki](#) who has developed his own Excel worksheet to calculate verification reports, the overall summary can be shown in a very easy-to-read format.

Anger.miki’s worksheet is the commonly seen results format seen on AVS Forum, with the dE Cake chart, Gamma Tracking graph, dE Summary, and much more.

Summary	
GS Points	10
Target Gamma	2.400
Avg Gamma	2.384
dE 2000	
Avg dE2000	0.2276
Max dE2000	0.8528
Avg GS dE2000	0.2273
Max GS dE2000	0.5392
0 - 0.499	960
0.5 - 0.999	40
1.0 - 1.5	0
> 1.5	0
Total points	1000

Anger.miki’s full Excel verification report worksheet is an outstanding piece of work and contains further graphs such as the commonly referred to dE Cake and Gamma Tracking graph as seen on AVSForum.

He has developed it much further since the creation of this guide, which has added other useful information and graphs, and it is constantly updated to reflect changes or updates that appear in ColourSpace software so will always be compatible with the latest software build.

Note: I have only shown the Summary by numbers here, the full sheet contains much more information.

Please contact [Anger.miki](#) directly via the link below if you are interested in having such a report generated for your calibration. Other services also available.

[Anger.miki Calibration Reports Service](#)

Final verification checks:

It is always advised to make some final Post-Calibration verification checks using verified accurate test patterns if possible, along with some real content.

As mentioned above, [Ted’s LightSpace CMS Calibration Disk](#) is perfect for this purpose and can be used either from your Blu-Ray Player (HDMI) using the disk, via USB using the digital files (MVK/MP4) or by using the same device used for patch generation with the recently released [PGenerator version](#) (HDMI).

All versions are 100% bit perfect and as accurate as can be.

Apply adjusted settings to All Inputs:

To ensure that the calibration has been applied accurately to all input sources, some settings will need to be manually transferred.

For generic settings such as OLED Light, Brightness, Contrast, Colour, Sharpness, you can just scroll to the bottom of the main Picture Mode Settings menu and click “Apply to All Inputs”.

For other settings within the “Expert Controls” and “Picture Options” sub-menus, you will need to manually copy these to each individual input. For example, if you have pre-calibrated the “Cool” Service Menu White Balance settings and used the corresponding setting in the user menu, then you will have to manually set this per each individual input. Likewise, settings such as “Super Resolution” in the Expert Controls sub-menu and “Noise Reduction” in the Picture Options sub-menu will also need to be manually transferred.

At this point, it is a personal preference as to whether you prefer to use certain image processing features. Motion Interpolation (TruMotion), Smooth Gradation and other such processing features are subjective choices that should be made on an individual basis. Nobody can tell you what looks best to you, this will be something you will have to experiment with on a personal level. It must be pointed out though, that turning on image processing can, and sometimes will, affect the final accuracy of any calibration performed so any adjustment made to these settings should be made knowing that this can happen.

Closing remarks:

There is nothing “magical” about the process described in this guide. Display Profiling and 3D LUT generation, as a rule, has a commonly standard approach.

However, each different display technology has its own different “quirks” that need to be accounted for and WRGB OLED is not immune to this. In fact, due to the instability of OLED in general, coupled with the WRGB sub pixel structure (White is not a combination of Red, Green and Blue), it requires a little extra work to produce the results seen above.

The amount of “man hours” invested in perfecting this specific process is beyond counting. A highly dedicated team of experienced enthusiasts have performed insane tests and experiments in order to achieve reference results and have then provided this knowledge to the community at large for free so that everyone has the chance to experience a perfectly calibrated display with minimal effort. This means more time enjoying the TV they purchased as opposed to wasting precious free time re-calibrating over and over trying to fix issues caused by botched calibrations using other (apparent “official”) methods.

As stated, the team has provided this process (and the “special” patch sequences”) to the community for free. They ask for, and expect nothing in return for the work put into this.

The only thing that we do ask is that you do not share the information provided in these guides publicly in open forums or social media platforms (*including, but not limited to such websites as YouTube*). The information is provided on a “good will” basis to be used by holders of a ColourSpace HTL licence or higher.

The team was put together, is managed and co-ordinated by Ted Aspiotis (AVSForum user ConnectEDDD). Without him, none of this would have been possible!!!

Ted dedicates much of his own personal free time assisting and supporting users all over the world, regardless of the software they choose to use or whether they are amateurs or professionals (*yes, he supports multiple Professionals the world over too*). He too asks for nothing in return.

In saying that, if you have found anything within this guide to benefit you and wish to show some appreciation, then I would ask that you please consider supporting Ted as he supports so many others.

[Please consider purchasing Ted’s LightSpace CMS Calibration Disk \(any version\)](#)

[Or possibly making a small donation to Ted to thank him for all he does to assist the calibration community](#)

Click the “Donate” button at the bottom of the page.

Finally, while much of the information contained within these guides is public knowledge and has already been posted throughout various forums on the internet, it has never before now been collated in such a way as to provide a simple to follow guide on LG OLED LUT calibration. The time invested in creating these guides and maintaining them as and when ColourSpace is updated, or new model LG’s are introduced, means it takes a lot of effort on my part.

The information is provided in good faith to users of ColourSpace software, and while some of the steps outlined can be of benefit to other software solutions, it is not intended to provide assistance or support for users of other software. I therefore request that information and steps outlined within the guides is not shared within the various forums or places such as YouTube.

Failure to respect this request will result in me removing access to the guides for any future users, and refusing to support and/or update the guides should the need arise.